ETH
Swiss Federal Institute of Technology Zurich

Seminar for Statistics

Department of Mathematics

Master Thesis Summer 2025

Anthony Eckmann

Managing Data Scarcity in Sports Analytics

Oversampling-Assisted Machine Learning for Forecasting Swiss Elite Climber Results

Submission Date: 01 September 2025

Advisor: Dr. Markus Kalisch

Acknowledgements

I am very thankful to my supervisor, Dr. Markus Kalisch, for always supporting me by giving the necessary guidance while leaving enough independence to pursue my own ideas. Furthermore I am grateful to my grandfather, Jean-Pierre, for careful corrections and thoughtful suggestions. I also thank my parents for everything. All remaining errors are my own.

Code and data availability

Due to confidentiality agreements with the Swiss Climbing Federation, the data is not publicly available. The code is available upon request via sekretariat(at)stat.math.ethz.ch.

Abstract

Data scarcity is a common challenge in competitive sports. Elite climbing exemplifies this constraint: athlete rosters are small and events are few, so even modest gains in predictive accuracy can affect outcomes. We study the prediction of Swiss elite climbers seasonal performance from yearly physical and psychological measurements. We construct a performance score by adapting the International Federation of Sport Climbing (IFSC) ranking to national and international events, and benchmark it against PageRank, Massey, Colley, and Elo; four popular ranking methods. The adapted IFSC score is most stable for our use case. To counter limited data, we implement two virtual sample generation (VSG) methods: Gaussian-mixture sampling for mixed continuous–categorical predictors and Bush-SMOTE that interpolates along Hamiltonian paths. Ten supervised learning models are fitted and evaluated on a held-out 2024 season dataset. Tree-based models benefit most from Bush-SMOTE: with Bush-SMOTE augmentation, gradient boosting reduces RMSE on the hold-out data by about 20% and raises $\boldsymbol{R}^2$ from 0.17 to 0.58, while XGBoost attains the highest $\boldsymbol{R}^2$ (0.62) on the same dataset. GMM-based augmentation yields notable RMSE reductions for distance-based SLMs (kNN, Ridge, SVM, Lasso, OLS). Our results show that data augmentation can improve predictive accuracy in sports analytics, where data tends to be limited and thus difficult to leverage. We discuss limits (model dependence, potential over-smoothing) and outline next steps: multi-season and cross-country data, learning-curve analyses to quantify when augmentation reaches its limits, and robustness checks for the performance score.

Contents

1 Introduction **1**

2 Performance Score **3**
2.1 Performance Evaluation . . . 3
2.2 Rating and Ranking . . . 3
2.3 Performance Metric Methods . . . 4
2.3.1 IFSC World Ranking Scoring System . . . 5
2.3.2 Google PageRank Algorithm . . . 8
2.3.3 Massey Method . . . 9
2.3.4 Colley Method . . . 10
2.3.5 Elo Rating System . . . 11
2.4 Comparing Methods . . . 12

3 Dataset Construction and Exploration **15**
3.1 Data Sources . . . 16
3.1.1 Physiological Tests . . . 16
3.1.2 Survey . . . 16
3.1.3 Performance Score . . . 17
3.2 Exploratory Data Analysis . . . 17
3.2.1 Performance Score Variable . . . 17
3.2.2 Dataset Visualization . . . 17
3.2.3 Variations based on gender . . . 21
3.2.4 Variations based on other categorical variables . . . 22
3.2.5 Relationships between variables . . . 23

4 Supervised Learning Models **25**
4.1 Challenges of small datasets . . . 25
4.2 Virtual Sample Generation (VSG) . . . 26
4.2.1 Gaussian Mixture Model VSG . . . 26
4.2.2 Adapted Bush Topology SMOTE . . . 30
4.3 Experimental Setup . . . 33
4.4 Comparison of Supervised Learning Models . . . 34
4.4.1 Linear Regression . . . 36
4.4.2 Ridge Regression . . . 37
4.4.3 Lasso Regression . . . 37
4.4.4 k-Nearest-Neighbors Regression . . . 37
4.4.5 Support Vector Regression with RBF Kernel . . . 38
4.4.6 Random Forests . . . 38
4.4.7 Gradient Boosting Machine (GBM) . . . 39
4.4.8 Extreme Gradient Boosting (XGBoost) . . . 40
4.4.9 Feed-Forward Neural Networks . . . 40
4.4.10 Bagged Neural Networks . . . 41

5 Conclusion and Future Work **43**

Bibliography **45**

A Data **49**
A.1 Variable Overview 49
A.1.1 Physiological Variables 49
A.1.2 Psychological Variables 50
A.1.3 Outcome Variable 50

Chapter 1

Introduction

In competitive sports, where resources are limited and performance margins are narrow, minor improvements in predictive capabilities can potentially enhance competitive outcomes (Lower and Czekanski, 2019). With the rise of data-driven decision making, the gap between data science and athletic performance optimization is narrowing. This thesis focuses on closing this gap within the context of competitive sport climbing, a discipline that has recently undergone rapid professionalization, particularly since its Olympic debut in 2020.

For small national federations such as the Swiss Climbing Federation, whose team counts around twenty elite athletes, accurate forecasting of athlete performance based on physical and physiological measurements can translate into noticeable performance gains on the competition circuit. Given the limited data availability due to such small teams and limited resources for regular collection of the athletes' physical and physiological attributes, statistical modeling presents a significant challenge. Data scarcity, heterogeneous competition conditions, and the continuous evolution of athletic requirements further complicate forecasting.

This work addresses these challenges directly, based on the Swiss Climbing Team's performance data from the 2024 season. The study introduces a comprehensive, season-long performance evaluation metric integrating results from both international and national competitions. This metric adapts the official scoring system from the International Federation of Sport Climbing (IFSC) (IFSC, 2025a), extending it systematically to national competitions. Additionally, we benchmark five established rating methodologies: the IFSC World Ranking system, Google's PageRank algorithm, the Massey method, the Colley method, and the Elo rating system. Through this comparative analysis, the study identifies the method providing the most stable and interpretable score trajectory for the 2024 season.

Even with a reliable seasonal evaluation metric, the challenge of limited data persists. To mitigate this, we propose two virtual sample generation (VSG) methods to artificially enlarge the training dataset, enhancing the predictive capabilities of supervised learning algorithms. The first method utilizes Gaussian mixture model (GMM) virtual sample generation, creating new, synthetic observations by fitting and sampling from Gaussian mixture distributions estimated from the original data. The second method, adapted Bush topology synthetic minority oversampling technique (Bush-SMOTE), generates new synthetic points strategically along selected Hamiltonian paths within the original data,

optimized to minimize validation error. Both methods, originally developed in materials science for addressing similar data limitations (Pais et al., 2024; Achar and Keith, 2024), demonstrate potential applications within sports analytics.

Finally, this thesis evaluates ten supervised learning algorithms, including linear regression, k-nearest neighbors, random forests, gradient boosting, and neural networks, comparing their predictive performance across three datasets: the base 2023 dataset containing the physical and physiological attributes of the athletes paired with the Performance Score, the base + GMM-VSG augmented dataset and the base + Bush-SMOTE augmented dataset. The results indicate that a decision tree based gradient boosting model trained on the base + Bush-SMOTE dataset performs best, reducing root mean squared error by approximately 20% and improving the coefficient of determination from 0.18 to 0.58.

The primary results of this thesis are therefore:

- Establish a reproducible and interpretable performance metric enabling the comparison of Swiss athletes within a season, accounting for both national and international results.
- Evaluate the effectiveness of GMM-VSG and Bush-SMOTE for predictive modeling in sports analytics, demonstrating their applicability in data-scarce environments.
- Develop an accurate predictive model capable of forecasting athlete performance using physiological and physical measurements collected by Swiss Climbing at the beginning of every competition season.

The thesis proceeds as follows. Chapter 2 details the construction and comparative evaluation of the proposed performance metrics. Chapter 3 provides a thorough description of the data sources and exploratory data analysis. Chapter 4 introduces the virtual sample generation techniques and compares supervised learning models to identify the most effective predictive model. Finally, Chapter 5 summarizes the findings, concluding with insights for future research directions.

Chapter 2

Performance Score

2.1 Performance Evaluation

Just like in most other competitive sports, the performance of bouldering and sport climbing athletes is evaluated and compared by holding competitions. There is an annual competition calendar which consists of world cups, continental cups, world championships and continental championships. In every competition, athletes are ranked according to their performance in boulder problems or on a lead route. Since not all competitions are equal, with some being more valuable than others, it is important to find a metric that summarizes an athlete's performance over an entire season. This section aims at proposing a metric that completes this task of comparing the athletes of the Swiss climbing team, considering all international and national competitions. To preserve anonymity, we associate a random athlete ID with every athlete, which will be used throughout this paper.

2.2 Rating and Ranking

Rating and ranking have been fundamental problems in statistics, with roots tracing back to the 13th century (Fidora and Sierra, 2011). Over time, various methods have been developed to assess and order entities, one of the most well-known being Google's PageRank algorithm, which ranks webpages based on their importance. In the mid-20th century, formal ranking systems were introduced especially for sports rankings, including the Elo rating system for chess and the Massey and Colley methods, used in Football. More recently, PageRank has been adapted for competitive rankings, utilizing network theory to assess performance and relative standing.

Most ranking methods are designed for pairwise comparisons, where two competitors face each other directly, such as in football matches or chess games. In climbing, however, the applicability of pairwise comparisons is less clear. Unlike head-to-head competitions, climbing events involve multiple athletes attempting the same lead route or set of boulder problems in a given round. Performance is then assessed based on measurable criteria, such as progress on a lead route or the number of tops and zones achieved in bouldering. This fundamental difference raises challenges in adapting traditional ranking models to accurately reflect athlete performance in climbing competitions. We will propose a method to overcome this problem in the next chapter.

2.3 Performance Metric Methods

We are going to compare different ranking methods to determine which is suited best for our performance metric. The following methods (or adapted versions of these methods) will be compared:

- IFSC World Ranking Scoring System
- Google PageRank Algorithm
- Massey Method
- Colley Method
- Elo Rating System

The first method is the method currently used by the International Federation of Sport Climbing, so we include it in our study. The latter four are among the best performing and most common methods currently used for sports rankings, which is why we include them (Langville and Meyer, 2012; Ochieng et al., 2022).

In order to adapt traditional ranking methods like the above to climbing competitions, we have to replace relative placements across an entire field of competitors with direct pairwise matches. For this, we will focus on an approach which creates pairwise comparisons between athletes who participated in the same competition, recording which athlete ranked higher in each case. Instead of a binary win/loss (higher/lower placement), we assign a weight based on the difference in ranking positions, as weightings are proposed to benefit the results according to Chartier et al. (2011).

We will apply exponential weighting to account for differences in athlete rankings. This approach ensures that weighting increases with rank difference but levels off beyond a certain point. The rationale behind this choice is that a sufficiently large rank gap indicates a significant performance disparity, such as one athlete reaching the finals or semifinals while another did not progress past qualifications. Since the number of athletes eliminated in qualifications is often large, and their performance differences may be minimal, we aim to prevent excessive weighting for minor variations in lower ranks while still capturing meaningful performance gaps at higher levels. We propose the following weighting function for pairwise comparisons:

$$w = (1 - \exp(-0.15 \cdot \text{rankdiff})) \cdot \text{stage} \ , \tag{2.3.0.1}$$

where:

rankdiff = absolute difference in rank of two athletes, compared pairwise

$$\text{stage} = \begin{cases} 1.0 & \text{At least one of the two athletes reached finals} \\ 0.8 & \text{Both athletes only made semifinals} \\ 0.8 & \text{One athlete made semifinals, one qualifications} \\ 0.2 & \text{Both athletes only made qualifications} \end{cases}$$

When looking at Figure 2.1, we clearly see that our pairwise comparison weighting function fulfills the above requests.

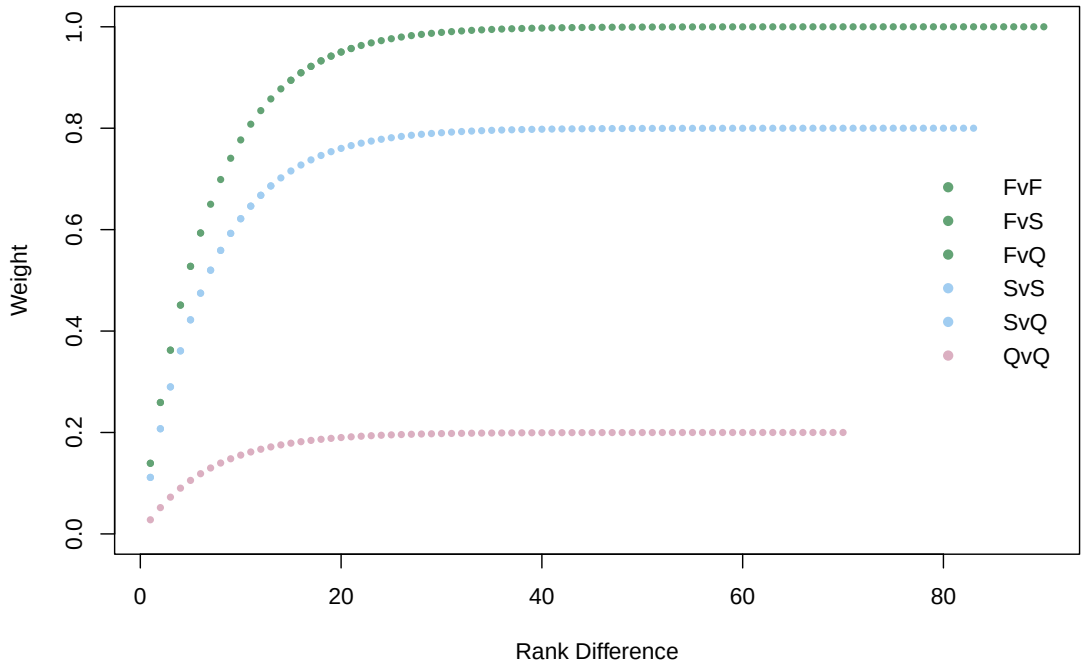


Figure 2.1: Weighting function for pairwise athlete comparison, where F stands for finals, S stands for semifinals and Q stands for qualification. For further details see the explanation of the stages above.

2.3.1 IFSC World Ranking Scoring System

We are going to adapt the yearly world ranking scoring system used by the International Federation of Sport Climbing described by IFSC (2025a,d). It is based on the standard competition ranking methodology (also known as SCR or "1224 method") commonly used in sports. The rankings are determined using a points-based system, where athletes accumulate scores from different international and continental competitions, such as IFSC World Cups and World Championships. This system helps identify top performers, track progression over time, and assess an athlete's standing on a global scale. Adapting this ranking system to this study allows for an evaluation of their performance within an international and national context.

Base points

Base points are the points that are obtained when finishing a competition at a certain rank. They are usually the same across all competitions in a sport and are determined by the ranking of the competition. In climbing, the IFSC also uses such a base points table, see Figure 2.2 (IFSC, 2025d). For this study, we will make use of the base point table by IFSC (2025d) to develop a metric useful for this case.

Rank	Points	Rank	Points	Rank	Points	Rank	Points
1st	1000	21st	145	41st	14	61st	4
2nd	805	22nd	130	42nd	13	62nd	4
3rd	690	23rd	120	43rd	12	63rd	4
4th	610	24th	105	44th	11	64th	3
5th	545	25th	95	45th	11	65th	3
6th	495	26th	84	46th	10	66th	3
7th	455	27th	73	47th	9	67th	3
8th	415	28th	63	48th	9	68th	3
9th	380	29th	56	49th	8	69th	2
10th	350	30th	48	50th	8	70th	2
11th	325	31st	42	51st	7	71st	2
12th	300	32nd	37	52nd	7	72nd	2
13th	280	33rd	33	53rd	7	73rd	2
14th	260	34th	30	54th	6	74th	2
15th	240	35th	27	55th	6	75th	1
16th	220	36th	24	56th	6	76th	1
17th	205	37th	21	57th	5	77th	1
18th	185	38th	19	58th	5	78th	1
19th	170	39th	17	59th	5	79th	1
20th	155	40th	15	60th	4	80th	1

Figure 2.2: Base points obtained at a competition given the final rank in a competition. The point distribution is determined by the IFSC in order to compute the IFSC world rank.

Weighting

Once the base points are determined, they are weighted depending on the type of competition and the location. This is done to emphasize the importance of certain competitions compared to others. For bouldering and sport climbing competitions, the weighting process is defined by IFSC (2025a,d). For simplicity, we will not take continental differences into account in this study. In addition, we will define weightings for national competitions in this metric, since national level competitions are not accounted for in the IFSC ranking. However, it makes sense in this case since we are only comparing Swiss athletes. The weighting is found in Table 2.1.

The weighting plays an important role in our performance evaluation. Below, we explain the reasoning behind the weights. All the weights except for the national level weights have been defined by the IFSC.

The World Championships are the most prestigious events in competition climbing and take place every two years. Since they did not occur in 2024, the year of this study, they are not included in our analysis.

The World Cups are the highest-level competitions in the circuit and are given full weighting (1.0). These events bring together the best athletes, many of whom focus exclusively on World Cups and hence do not compete in lower-level events, such as European Cups, during the season. Since World Cups are the main benchmark for elite performance, they are the most relevant competitions for this study.

The Continental Championships, in this case, the European Championships, are the second most important competitions. They are held annually and serve as good preparation for World Cups while also helping athletes gain recognition within their teams and among sponsors. Since the IFSC assigns them a weighting of only 0.2, many athletes tend to prioritize World Cups over Continental Championships. Yet there are still quite a few World Cup level athletes who partake in Continental Championships.

The Continental Cups, specifically the European Cups in this study, have the lowest weighting in the competition circuit, set at 0.1. These events primarily serve as opportunities for athletes who do not participate in World Cups. For athletes competing in World Cups, there is little reason to participate in Continental Cups, given their low weighting. However, good performance in continental cups can result in athletes being allowed to participate in World Cups.

Side remark: there are strict athlete quotas per country for World Cups which are determined by IFSC (2025b). It is up to the country's climbing federation to decide which athletes are allowed to participate in the World Cups. Here are the summarized rules:

- Guaranteed fixed quotas of two athletes per gender/discipline/Country
- Variable quotas, up to a maximum of four athletes per gender/discipline/Country, depending on performance (performance is measured by a country's number of athletes in the top 40 world rankings)
- World Cup host country quotas, four athletes per gender/discipline/Country

The National Cups, specifically the Swiss Climbing Cups, are not considered in the IFSC world rankings. However, since this study focuses on Swiss athletes, it is relevant to include them in our analysis. Given their lack of international significance, we assign them a low weighting of 0.01. Despite this, they hold value within the Swiss Climbing Federation, particularly for selecting upcoming athletes for the Swiss Climbing Team. As a result, these competitions are essential for emerging athletes aiming to progress to the international circuit. Additionally, they serve as valuable preparation for higher-level competitions.

It is important to recognize that an athlete's performance at the international level does not always correspond to their results in national competitions like the Swiss Climbing Cups. This discrepancy arises because these events hold less importance for elite athletes, who focus primarily on performing well in top-tier international competitions. As a result, national-level competitions are often approached as training opportunities or, in some cases, as mandatory events.

The Swiss Championships, however, differ from the Swiss Climbing Cups in that they hold a certain level of prestige. While they do not contribute to international rankings, a strong performance in the Swiss Championships can enhance an athlete's reputation within the Swiss Climbing Federation and increase their visibility among national sponsors, making them an important event at the national level. Therefore we assign them a weighting of 0.1.

Competition	Weight
World Cup	1
Continental Cup	0.1
Continental Championship	0.2
Swiss Climbing Cup	0.01
Swiss Championship	0.1

Table 2.1: Weights for the different competitions

Computation of the score

Once the points are determined and weighted, they can be applied to all competition results of the athletes. The IFSC ranks the athletes according to the sum of their six best competition results of a season, competition result being defined as weighted points (IFSC, 2025a). We will adapt this method slightly and give some additional constraints. Three of the results shall be from international or continental competitions such as world cups or European cups. The three other results shall be from national competitions, where at most one of the results shall be from the Swiss Championships. For simplicity, we will not include the competitions held as part of the Olympic Games of 2024.

The formula used to calculate the performance score of an athlete is given as follows:

$$P = \sum_{b \in B} b + \sum_{d \in D} d \,, \quad \text{where} \quad B = \underset{A' \subset A, |A'| \leq 3}{\operatorname{argmax}} \Sigma_{a \in A'} a \quad \text{and} \quad D = \underset{C' \subset C, |C'| \leq 3}{\operatorname{argmax}} \Sigma_{c \in C'} c \,,$$

where:

- $A = \left\{ a \;\middle|\; a = \begin{array}{l} \text{weighted points for a given international/continental} \\ \text{competition in 2024 for a specific athlete} \end{array} \right\}$
- $C = \left\{ c \;\middle|\; c = \begin{array}{l} \text{weighted points for a given national} \\ \text{competition in 2024 for a specific athlete} \end{array} \right\}$
- C' holds at most one result of a Swiss Championship competition
- P is the performance score of the specific athlete

The scores obtained for the 2024 season can be found in Table 2.2.

2.3.2 Google PageRank Algorithm

The PageRank algorithm, originally developed by Google to rank web pages based on their importance, has been widely adapted to evaluate athlete performance across various sports. This makes it a promising approach for assessing climbing performance.

The algorithm operates by representing the World Wide Web as a directed graph, where nodes correspond to web pages and directed edges represent hyperlinks between them. Given the immense scale of the web - Yahoo reported indexing 19.2 billion pages in 2005 (Markoff, 2005) - this graph is highly complex and densely connected. The algorithm assigns an initial score to each page and iteratively updates it based on the scores of linked pages. A page receives a higher rank if it is linked to by many important pages. The score update follows a probability-based formula, where a "random surfer" either follows a link with a certain probability or jumps to a random page. This process eventually converges,

producing a ranking that reflects the relative importance of each page in the network. Further details on the PageRank algorithm can be found in Langville and Meyer (2006).

For the comparison of athletes sport climbing and bouldering performance, we construct a network where directed edges represent ranking comparisons between athletes in the same competition instead of hyperlinks between web pages. An athlete who consistently ranks higher than other competitors will receive a higher score, similar to how a webpage linked by authoritative sources gains more weight. This method allows us to incorporate indirect performance relationships, capturing the competitive structure beyond direct pairwise comparisons.

Weighting

For weighting, we are going to consider following factors:

- Ranking Difference Weight
- Competition Type Weight

To account for ranking differences, we apply the weighting function defined in Equation 2.3.0.1, ensuring that pairwise comparisons reflect both the rank disparity and competition stage of the athletes.

Additionally, competition type weights are assigned according to the values presented in Table 2.1.

The final weights used in the PageRank algorithm are obtained by multiplying these two weighting factors, integrating both rank-based and competition-specific adjustments. They are implemented by defining the `weights` argument in the `page_rank` function from the `igraph package`.

Computation of the score

Given all the information above, we can run the PageRank algorithm and determine a performance score for each athlete for the 2024 season. The obtained scores are found in Table 2.2.

2.3.3 Massey Method

The Massey method, introduced for ranking sports teams based on pairwise game outcomes, provides a linear algebra-based approach to evaluating competitive performance. The method constructs a system of linear equations that models the differences in team (or athlete) rankings as a function of observed performance margins. By treating competition results as a network of pairwise comparisons, the Massey method determines rankings by minimizing the sum of squared differences between observed rank margins and computed rankings, ensuring a least-squares optimal solution. More information on the method can be found in Massey (1997).

We adapt the Massey method to the context of climbing performance analysis, where multiple athletes compete simultaneously rather than in head-to-head match-ups. Instead of win-loss outcomes, we use rank differences to define performance comparisons. This adaptation enables the method to capture performance differences across varying levels of competition, potentially allowing for a more robust ranking system that accounts for both direct and indirect competition between athletes.

Weighting

For weighting, we are going to consider following factors:

- Ranking Difference Weight
- Competition Type Weight

To account for ranking differences, we apply the weighting function defined in Equation 2.3.0.1, ensuring that pairwise comparisons reflect both the rank disparity and competition stage of the athletes.

Additionally, competition type weights are assigned according to the values presented in Table 2.1.

The final weights used in the Massey method are obtained by multiplying these two weighting factors, integrating both rank-based and competition-specific adjustments. They are implemented by weighting the vector b in $M \cdot \hat{s} = b$, where M is the Massey matrix, $\hat{s}$ is the performance score and b is the total point differential of an athlete (in this case the total weight differential).

Computation of the score

Given all the information above, we can run the Massey method and determine a performance score for each athlete for the 2024 season. The obtained scores are found in Table 2.2.

2.3.4 Colley Method

The Colley method, developed for ranking sports teams based on win-loss records, provides a probabilistic approach to ranking competitors by constructing a system of linear equations. Unlike purely margin-based methods, the Colley method incorporates a bias toward the mean, ensuring that rankings remain well-defined even with limited data. It operates by constructing a modified Laplacian matrix that encodes the number of competitions each athlete has participated in, as well as a score vector reflecting relative performance. The system is then solved to obtain rankings that are robust to variability in competition frequency and match-ups. More can be found in Colley (2002).

We modify the Colley method to suit the ranking structure of climbing competitions, where athletes compete in the same event rather than in direct match-ups. Since traditional win-loss records do not apply, we construct a ranking system based on relative placements within each competition. To enhance the method's sensitivity to performance differences, we will integrate a weighting scheme. This adaptation ensures that rankings remain fair and robust, even when athletes compete in different numbers of events or face varying levels of competition.

Weighting

For weighting, we are going to consider following factors:

- Ranking Difference Weight
- Competition Type Weight

To account for ranking differences, we apply the weighting function defined in Equation 2.3.0.1, ensuring that pairwise comparisons reflect both the rank disparity and competition stage of the athletes.

Additionally, competition type weights are assigned according to the values presented in Table 2.1.

The final weights used in the Colley method are obtained by multiplying these two weighting factors, integrating both rank-based and competition-specific adjustments. They are implemented by weighting the matrix C and the vector b in $C \cdot \hat{s} = b$. C is the Colley matrix, $\hat{s}$ is the performance score and b is given by $b = 1 + \text{total wins} - \frac{\text{total games}}{2}$ (wins and games are weighted).

Computation of the score

Given all the information above, we can run the Colley method and determine a performance score for each athlete for the 2024 season. The obtained scores are found in Table 2.2.

2.3.5 Elo Rating System

The Elo rating system, developed for ranking and rating chess players, is a dynamic method that updates an athlete's rating based on performance relative to expectations. Unlike the previous ranking methods, the Elo rating system adjusts ratings iteratively after each competition, increasing a competitor's score when they outperform expectations and decreasing it when they underperform. The magnitude of rating adjustments depends on the difference in initial ratings between competitors, ensuring that defeating a highly ranked opponent results in a greater rating gain than defeating a lower-ranked one. Further detail is found in Elo (1978).

To adapt Elo for climbing competitions, where multiple athletes compete simultaneously rather than in head-to-head matches, we construct a framework based on relative rankings within each event. Pairwise comparisons are derived from final placements, with a weighting scheme that considers the rank difference of the two athletes. Additionally, a smoothing factor is introduced to prevent excessive rating fluctuations due to outlier performances. This approach allows for an evolving ranking system that continuously integrates new competition results while maintaining stability over time.

Weighting

For weighting, we are going to consider following factors:

- Ranking Difference Weight
- Competition Type Weight

To account for ranking differences, we apply the weighting function defined in Equation 2.3.0.1, ensuring that pairwise comparisons reflect both the rank disparity and competition stage of the athletes.

Additionally, competition type weights are assigned according to the values presented in Table 2.1.

Athlete ID	PageRank	MasseyRating	ColleyRating	Elo	IFSC Ranking
1	0.031	-0.018	0.400	0.203	53.650
2	0.026	0.003	0.500	0.504	71.000
3	0.323	-0.013	0.487	0.328	347.600
4	0.334	0.028	0.590	0.861	759.650
5	0.097	-0.007	0.476	0.417	210.600
6	0.084	-0.005	0.484	0.458	134.800
7	0.428	0.222	0.804	4.909	1486.100
8	0.026	-0.027	0.407	0.261	71.600
9	0.022	0.004	0.500	0.502	18.500
10	0.026	-0.170	0.257	-1.710	127.700
11	0.182	0.036	0.584	1.096	947.400
12	0.277	0.059	0.603	0.543	468.100
13	0.039	-0.125	0.345	-2.101	444.450
14	0.105	0.012	0.564	0.730	193.550

Table 2.2: Ranking Methods Summary

The final weights used in the Elo Rating System are obtained by multiplying these two weighting factors, integrating both rank-based and competition-specific adjustments. They are implemented by setting the argument `k()` of the `elo.run` function of the `elo package` to our weighting function. We also set the initial elo to 0.5 for all athletes.

Computation of the score

Given all the information above, we can run the Elo Rating System and determine a performance score for each athlete for the 2024 season. The obtained scores are found in Table 2.2.

2.4 Comparing Methods

After deriving a performance score for each athlete using the five different ranking methods, IFSC Ranking System, PageRank, Massey, Colley and Elo, we now proceed to compare these methods to determine which best aligns with our objectives.

To compare the different ranking methods, we are normalizing the scores of each method. For this, we are using the common z-score normalization which is given by $z = \frac{x-\mu}{\sigma}$, where $x =$ value, $\mu =$ mean and $\sigma =$ standard deviation.

In Figure 2.3 and Figure 2.4 we see the different methods, for female and for male athletes.

Overall, the ranking methods produce similar results, with the highest and lowest performing athletes generally remaining consistent across methods. The Massey and Colley methods yield comparable scores, possibly since both rely on a least squares approach. The PageRank algorithm seems to emphasize top-performing athletes while assigning significantly lower scores to those with weaker performances. Additionally, top performing athletes tend to compete in more events than lower ranked athletes. This pattern aligns with PageRank's original purpose - ranking web pages - where it is essential to distinguish between "better" and "worse" pages and identify those that are most frequently cited. In

our context, this corresponds to the weighted number of victories an athlete achieves. Interestingly, it seems as if PageRank produces results most similar to the adapted IFSC Ranking method. This is likely because both approaches reward athletes who compete more frequently with higher point accumulation.

For further analysis, we will proceed with the adapted IFSC Ranking method, as it seems to offers a well-balanced approach and it is not based on pairwise comparison, which makes it easier to implement and less computationally expensive, regardless of other athletes performances. Furthermore, it is easily applicable to all competing athletes, regardless of nation. This might be beneficial in further analysis.

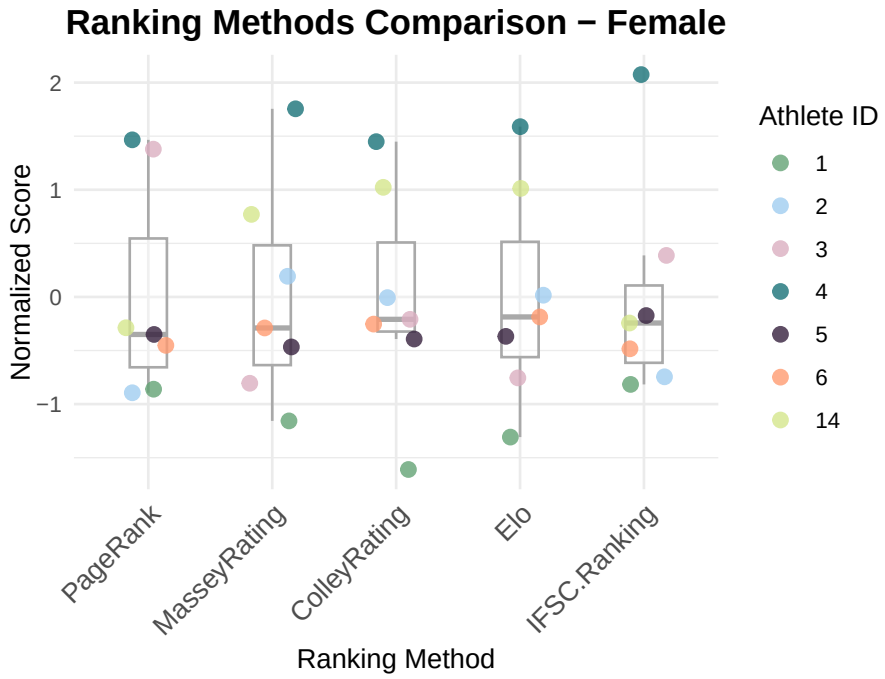


Figure 2.3: Female Athletes Ranking Methods Comparison

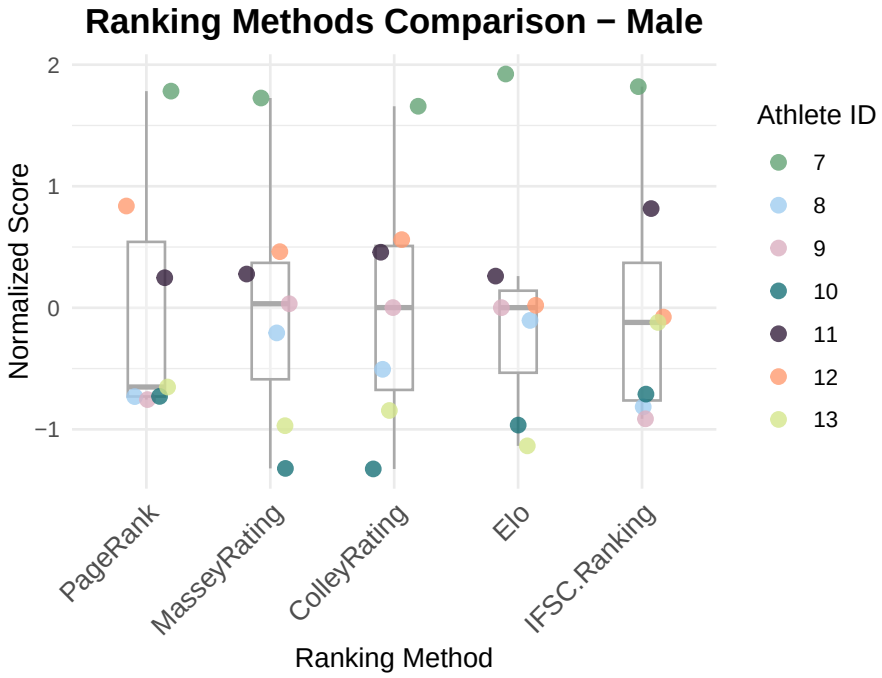


Figure 2.4: Male Athletes Ranking Methods Comparison

Chapter 3

Dataset Construction and Exploration

In order to compare different supervised learning models, we will use datasets that consist of both objective performance metrics and subjective psychological assessments of the Swiss national climbing team. The goal is to identify a model that can effectively quantify athletic performance based on the available data. Additionally, the findings could have practical implications for optimizing training methodologies, injury prevention strategies, and mental performance coaching in competition climbing. The datasets consists of three main components:

- Physiological measurements, categorized into strength, flexibility, and coordination. These assessments were conducted under controlled conditions, ensuring consistency across all athletes. All tests were performed on the same day at the Swiss climbing training facility, minimizing external influences.

- An athlete survey, in which the same group of athletes reflected on their 2024 competition season. The survey was designed to capture psychological and emotional factors that could influence performance in climbing competitions. Athletes provided self-reported insights into their mental state during competition, training habits, relationships with teammates and coaches, and dietary habits, among other subjective experiences.

- The outcome variable, which is a performance score computed for each athlete based their performance.

By combining the above, the datasets provide an opportunity to explore how physiological and psychological variables interact in the context of competition climbing. Given that the number of athletes is relatively small, the quality and consistency of the dataset are of great importance to ensure meaningful insights can be drawn from the analysis.

In Table 3.1, a overview of the available datasets are is given. Furthermore, in Appendix A.1, all variables appearing in the thesis are described.

	Physio. Variables	Psycho. Variables	Outcome
2023	Yes	No	Yes
2024	Yes	No	Yes
2025	No	Yes[1]	No

Table 3.1: Dataset overview

3.1 Data Sources

3.1.1 Physiological Tests

At the beginning of each year, the Swiss Climbing Team gathers to plan and discuss the upcoming competition season. As part of this meeting, athletes undergo a series of physiological tests designed to assess strength, flexibility, and coordination. These tests allow coaches and officials to track individual progress over time and compare physiological attributes within the team.

To ensure consistency and comparability, the tests are conducted in the same facility, following the same procedures and order each year. The assessment begins with strength and coordination tests, including various finger strength measurements and the SLAP test. This is followed by flexibility tests, such as splits and foot raises. Finally, the testing session concludes with another strength assessment, specifically the one-repetition max weighted pull-up. All data is systematically collected and recorded by Swiss Climbing officials, ensuring accuracy and reliability in the dataset. This structured approach minimizes variability and maintains a high level of data quality.

3.1.2 Survey

We specifically designed a survey for this study, as no similar surveys had been conducted in previous years. As a result, there is no other survey data available for direct comparison. The goal was to gather athletes' reflections on their 2024 competition season, capturing insights that are not measured through physical tests.

The survey was conducted in January 2025, on the same day as the annual Swiss Climbing gathering and physiological testing, through an online questionnaire. To ensure clarity, athletes completed the survey between test sessions and had the opportunity to ask questions if anything was unclear.

The survey consists of four distinct sections, which are:

Training and Competition: *Questions about training habits, competition habits, mental aspect of competitions etc.*

Daily Routine: *Questions about sleep, nutrition etc.*

Swiss Climbing Team: *Questions about the team in general, about coaches etc.*

Goals: *Questions about specific goals for the season*

[1] The survey was conducted with the athletes of 2024 to reflect on the 2024 season.

3.1.3 Performance Score

The performance score is based on individual athletes competition rankings of the 2024 season. The competition rankings were collected from the IFSC results webpage (IFSC, 2024) and the SAC results webpage (SAC-CAS, 2024). The performance score is described in Chapter 2.

3.2 Exploratory Data Analysis

Given the very small sample size of the dataset, conducting exploratory data analysis (EDA) is a crucial step in ensuring meaningful insights before applying supervised learning models. When working with a limited number of data points, small variations can have a significant impact on the results. Additionally, it plays a key role in feature selection, ensuring that supervised learning models focus on the most relevant attributes.

Before conducting EDA, we combine the physiological, psychological, and performance score variables into a single dataset. This merged dataset serves as the final version that will be used for all further analysis and model development.

We will define several objectives in order to maximize the information gained through exploratory data analysis.

- Summary of the performance score variable
- Physiological variables
- Variations based on gender
- Variations based on other categorical variables
- Relationships between variables

3.2.1 Performance Score Variable

We have 14 observations of the performance score variable, with the lowest recorded value at 18.5 and the highest at 1486.1. Theoretically, scores could range from 0, representing athletes who did not compete, to approximately 3500. The observed mean is 381.05, while the median is 202.07. Examining the density of the variable (see Figure 3.1), we observe a right-skewed distribution, with most values falling within the 0 to 500 range. This distribution is expected, as Switzerland's climbing team lags behind top competition climbing nations such as France, Japan, and the USA. According to the IFSC Team World Rankings, Switzerland ranks 14th in the women's category and 12th in the men's category (IFSC, 2025c).

3.2.2 Dataset Visualization

Let us examine the different types of variables present. Figure 3.2 provides a summary of various athlete attributes, with the red line indicating the mean for each variable. Most athletes train approximately five days per week, with no one training fewer than four or more than six days. Sleep also appears to be a crucial factor, as no athlete reports sleeping less than six hours on a regular basis or less than seven hours before a competition. The majority sleep around eight hours normally and nine hours before competing. Additionally,

the average height is approximately 170 cm, while the average body weight is around 67 kg.

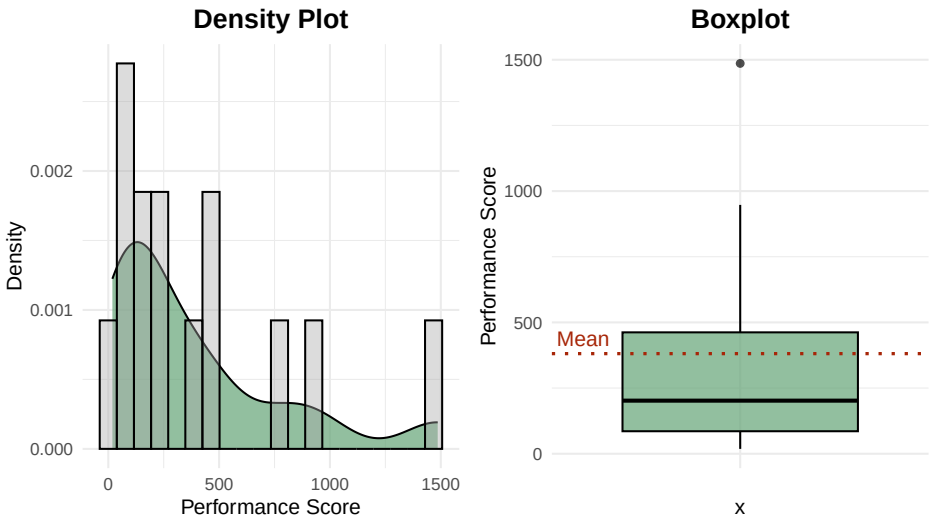


Figure 3.1: Density plot and box plot of the performance score

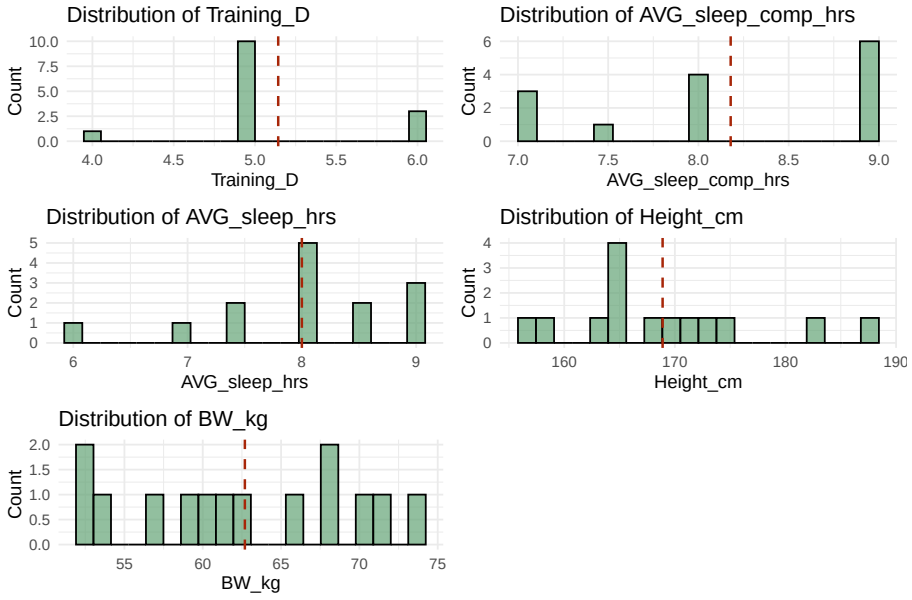


Figure 3.2: Distribution of athlete attributes with the mean displayed as dotted red line.

In Figure 3.3, we observe that the performance score tends to increase with greater finger strength relative to body weight (FS_MVC_L and FS_MVC_R). Similarly, relative pull-up strength (PullUp_MW) and the Slap test (Slap_L and Slap_R), which measures how high an athlete can dynamically pull up in a single motion, also appear to be positively associated with performance score. However, there are some potential outliers, which could be attributed to injuries sustained throughout the season. The remaining variables primarily assess flexibility, and no clear trends emerge from the scatter plots.

When examining some psychological variables assessing athletes well-being (Figure 3.4),

no obvious correlations with performance score are apparent at first glance, which is unexpected. However, there may be gender-specific differences which will be analyzed in the next section.

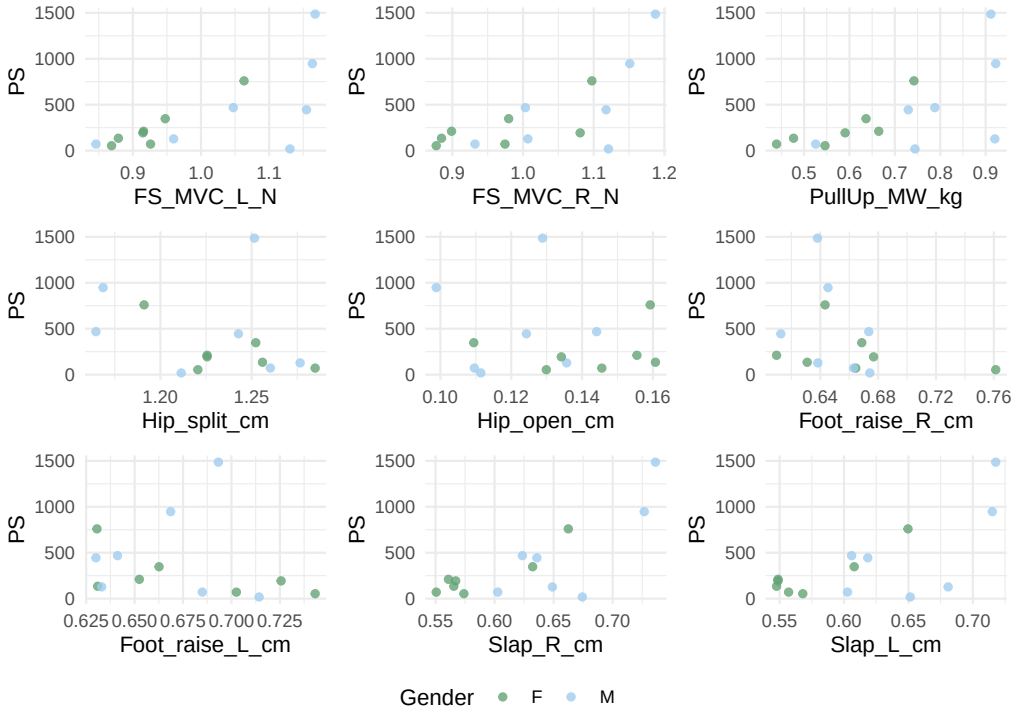


Figure 3.3: Scatterplot with physiological variables on the x axis and the performance score on the y axis.

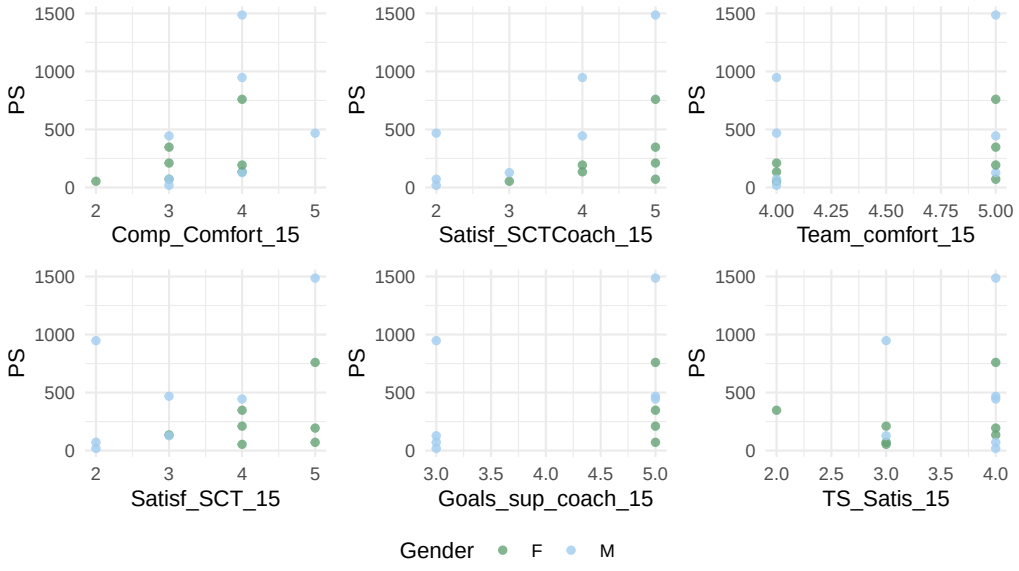


Figure 3.4: Scatterplot with psychological variables (values ranging from 1 being the worst to 5 being the best) on the x axis and the performance score on the y axis.

In Figure 3.5, the distributions of the categorical variables are displayed. As expected,

we observe that a greater number of athletes follow a specific diet (Diet). Similarly, more athletes reported having specific goals for the season compared to those who did not (Goals). Additionally, the majority of athletes had external coaches as their primary coaches (Main_coach). Athlete preferences between training and competition appear to be fairly balanced (Preference_TCB). Likewise, the self-reported performance in competitions versus training sessions shows no strong skew in either direction (Performance_better).

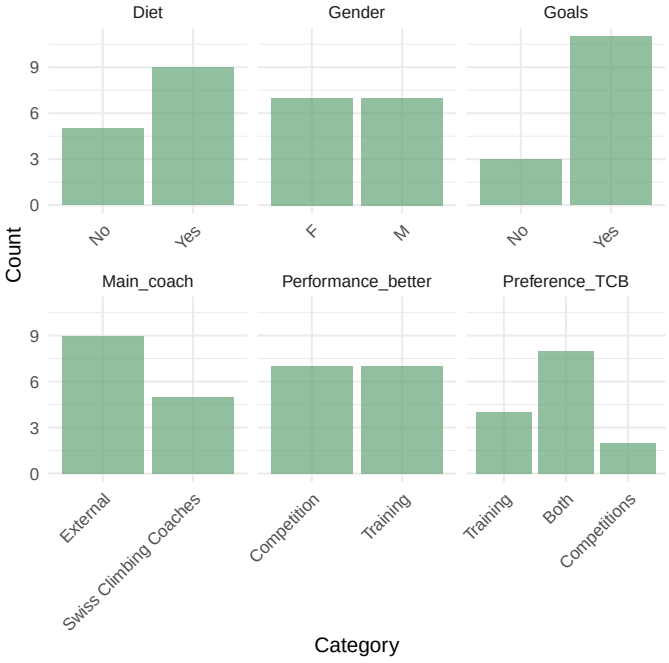


Figure 3.5: Barplots of the categorical variables

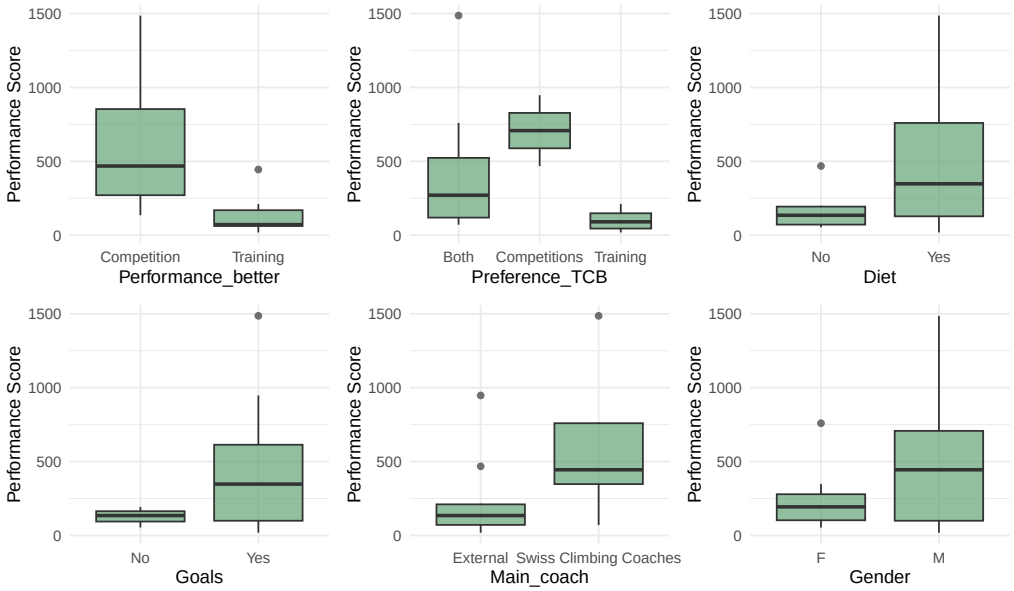


Figure 3.6: Box plots of the categorical variables for the performance score

When comparing the performance score across different categories (Figure 3.6), we observe that athletes who reported performing better in competitions (Performance_better) tend to have a higher performance score, which makes sense. Similarly, athletes who prefer competitions over training (Preference_TCB) also appear to achieve higher performance scores. Athletes who had specific goals seem to potentially have higher performance scores, however this needs further inspection. The same holds for athletes having a diet and athletes whose main coaches are the ones from the Swiss Climbing team.

3.2.3 Variations based on gender

As one would expect, examining Figure 3.7, we observe a clear trend indicating that male athletes generally possess greater relative strength compared to female athletes (FS_MVC, PullUp_MW, Slap). In contrast, flexibility-related variables appear to be more balanced, with the exception of hip mobility, where female athletes demonstrate a greater ability to open their hips (Hip_open).

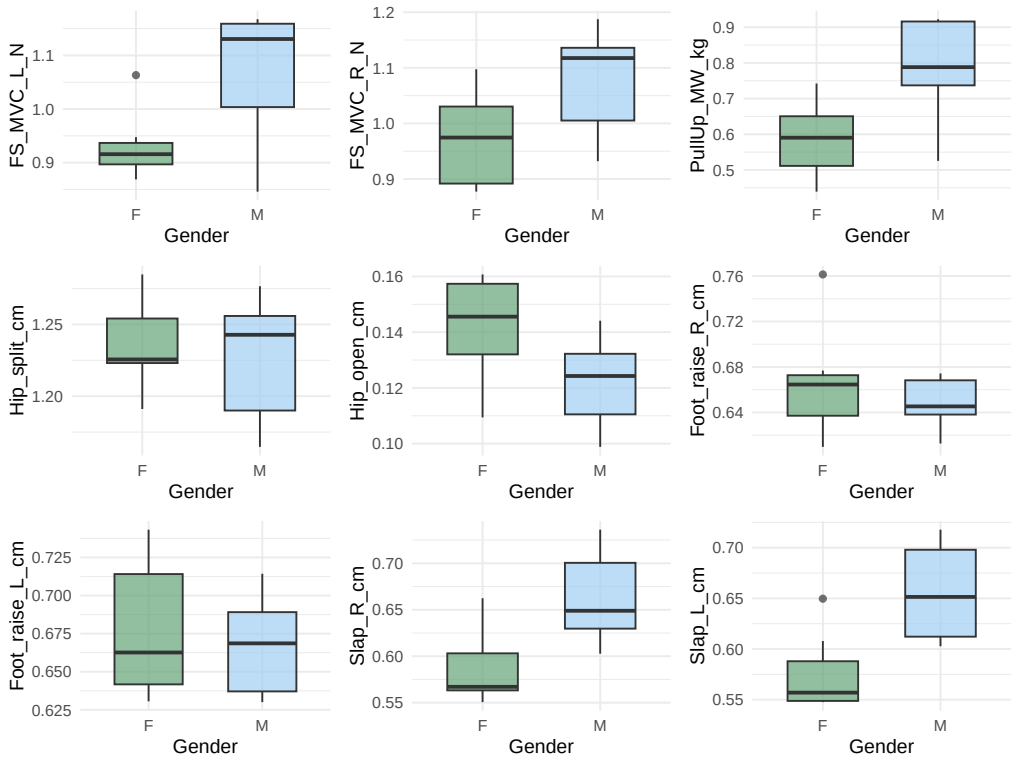


Figure 3.7: Box plots of the physiological variables by gender

For the psychological variables, we note several interesting things by assessing Figure 3.8. Male athletes seem to feel more comfortable during competitions than female athletes (Comp_Comfort). Furthermore, they appear to be less satisfied with the Swiss Climbing Team coaches compared to female athletes (Satisf_SCTCoach). A similar trend is observed in overall satisfaction with the Swiss Climbing Team (Satisf_SCT). Among those athletes who had specific goals for the season, male athletes reported feeling less supported by the SCT coaches in achieving these goals compared to their female counterparts (Goals_sup_coach).

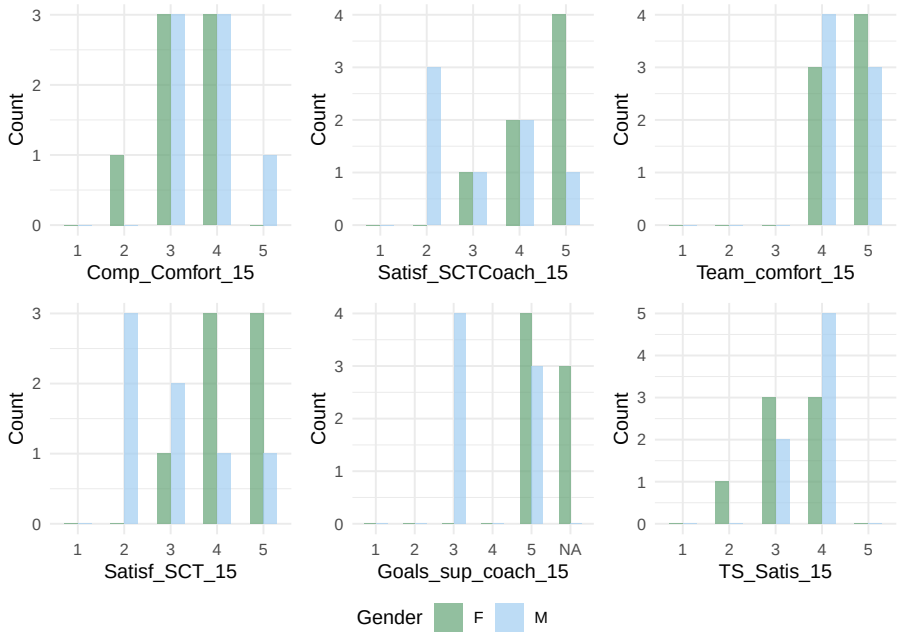


Figure 3.8: Bar plots of the psychological variables (values ranging from 1 being the worst to 5 being the best) by gender.

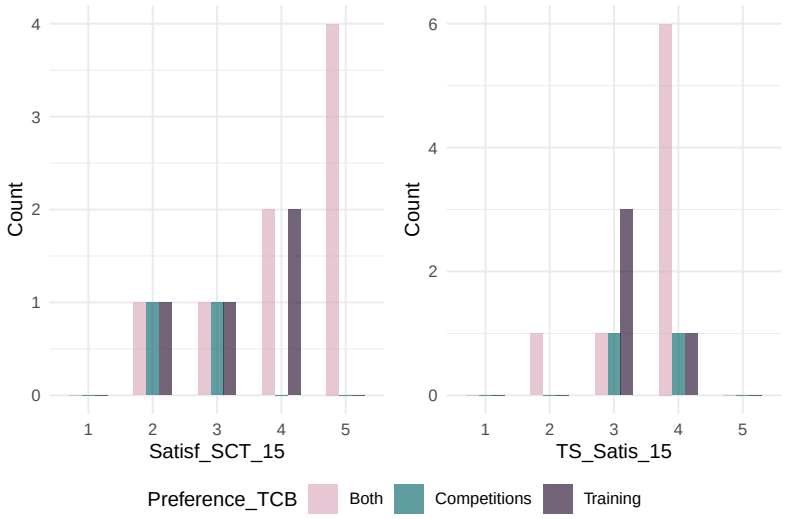


Figure 3.9: Bar plots of the general satisfaction with the SCT and the personal satisfaction after a training session given the training /competition preference.

3.2.4 Variations based on other categorical variables

For athletes who follow a diet, no noticeable categorical differences are observed in either physiological or psychological variables. This is unsurprising for psychological factors, as dietary habits are unlikely to influence aspects such as satisfaction with the Swiss Climbing Team or overall team dynamics. While one might argue that adhering to a diet could contribute to a greater sense of well-being, it is also possible that athletes who do not follow a diet already feel equally well and therefore see no need for one. Regarding physiological variables, no significant differences were found based on diet. This is un-

expected for strength-related measures, as a diet specifically structured for climbing and sports is known to enhance training effectiveness (Chmielewska and Regulska-Ilow, 2023; Tan, 2025). However, our categorical variable only distinguishes between having a diet or not, without accounting for variations in dietary definition among athletes. Differences in individual interpretations of what constitutes a "diet" could explain these findings.

When comparing athletes based on their preference for training versus competitions, two interesting patterns emerge (Figure 3.9). Athletes who enjoy both training and competitions equally report the highest overall satisfaction with the Swiss Climbing Team. Conversely, athletes who prefer competitions tend to be less satisfied with the SCT. A possible explanation is that the Swiss Climbing Team may have areas for improvement in aspects related to competitions, such as organization, administration, and competition support. Additionally, athletes who equally enjoy both training and competitions report the highest levels of satisfaction following training sessions.

3.2.5 Relationships between variables

Looking at a correlation plot (Figure 3.10), we notice strong correlations among the various strength-related variables, suggesting that physically stronger athletes tend to have more strength generally, rather than excelling in just a single specific measure. A strong negative correlation between training days (Training_D) and full rest days (FullRest_D) is also observed, which is expected, as more training days naturally leave fewer days for full rest. Interestingly, we also observe a correlation between the extent to which an athlete's life revolves around climbing (Climbing_Life) and their reported frustration during training sessions (TS_Frust). This could indicate that athletes who are more invested in climbing may place higher expectations on themselves, potentially leading to greater frustration when performance falls short. Another notable pattern is the correlation between goal-related and satisfaction variables (Goals, Satisf) and physical characteristics such as height and body weight. While this may initially seem unusual, it is likely driven by gender differences that are inherently reflected in both body metrics and possibly in responses to coaching or support. A similar gender-related effect likely explains the correlation between hip flexibility (Hip_open) and height/body weight. Lastly, we notice a correlation between how much climbing influences one's lifestyle (Climbing_Life) and the ability to lift the feet—though this appears to be an isolated case and is likely due to noise in the data rather than a meaningful relationship.

From the tornado plot of the Spearman correlation with the performance score (Figure 3.11), we observe that strength-related variables such as FS_MVC, PullUp_MW, and Slap show strong positive correlations with performance. The variable Goals_achieve, reflecting how well athletes felt they achieved their season goals, shows a strong correlation with performance as well, which is to be expected, since higher-performing athletes are more likely to have reached their objectives. Similarly, Comp_Comfort, indicating how comfortable an athlete feels during competitions, is also strongly correlated with performance, further underscoring the importance of mental readiness in high-level performance. Other potential correlations include Climbing_Life (how much of an athlete's life revolves around climbing), AVG_sleep (average sleep duration), satisfaction with team coaches (Satisf_SCTCoach), and overall comfort within the team (Team_comfort). These suggest that both lifestyle and psychological well-being may contribute to performance outcomes. Finally, we again observe a negative correlation with the ability to raise the feet, though this likely stems from random variation or noise rather than a true underlying relationship.

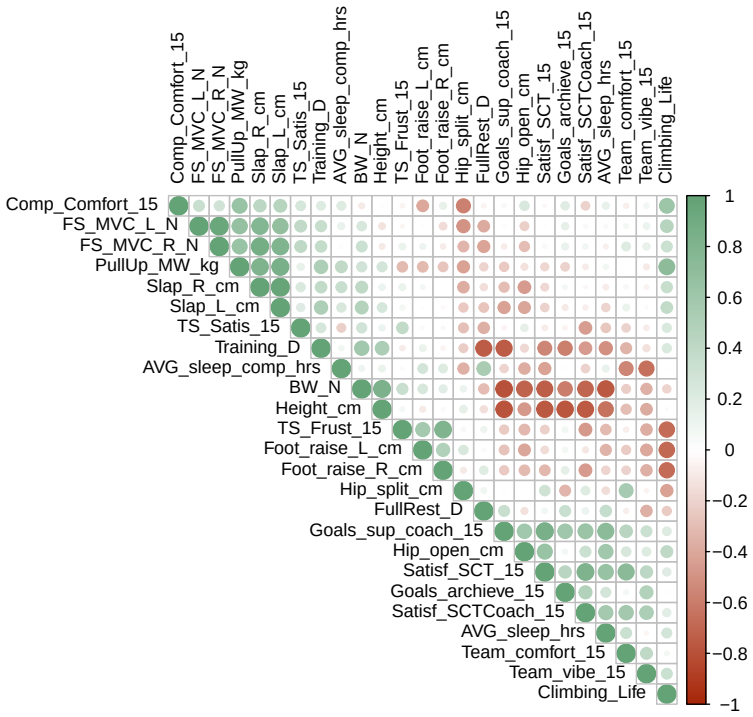


Figure 3.10: Correlation plot of all numerical variables

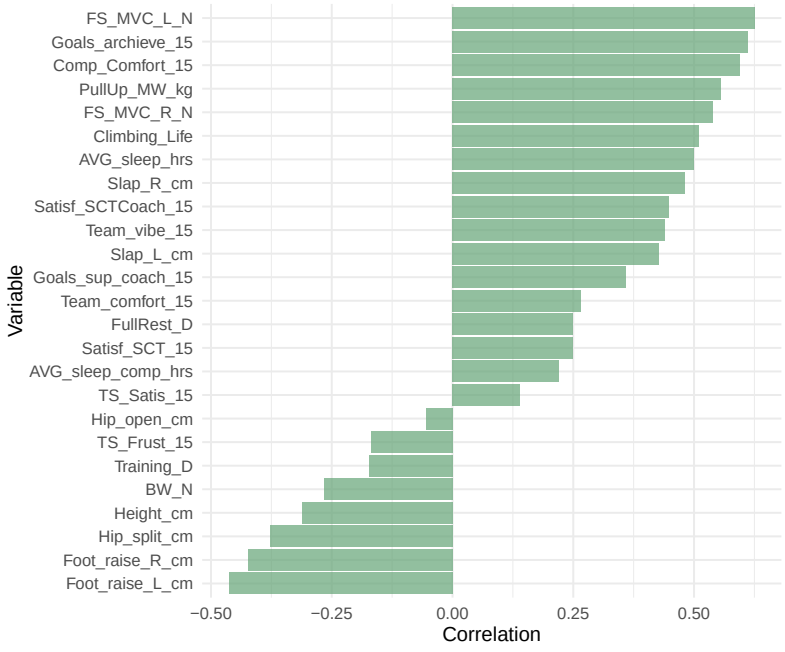


Figure 3.11: Tornado plot of the Spearman correlation of every numerical variable with the performance score.

Chapter 4

Supervised Learning Models

In recent years, machine learning models have become essential tools for analyzing and predicting data, with applications across many fields, from financial forecasting to medical diagnostics (Jordan and Mitchell, 2015). While advances in digitalization have led to increasingly large and diverse datasets in many areas (Fan and Bifet, 2013), many real-world domains continue to face the challenge of limited data availability. This is particularly true in specialized and niche fields, where data is limited due to the small number of individuals involved and the complexity and cost of collecting high-quality measurements. Specifically in elite sports, it is challenging due to small athlete populations and the competitive, high-stakes nature of the environment.

In this chapter, we shift our focus to the core of this thesis, the comparison of supervised learning algorithms applied to the small (according to Gong et al. (2017)) dataset of the Swiss National Climbing Team. After having introduced a performance score in order to compare athlete's performance and having explored athlete characteristics in detail in the previous sections, we now examine how different algorithms perform in predicting the performance score from physiological and psychological attributes. Accurately predicting athlete performance based on those attributes could support coaches and sports scientists in making informed, data-driven decisions. For example, it could aid in identifying which physiological or psychological factors to prioritize in training, or recognizing athletes with high potential early on.

4.1 Challenges of small datasets

Working with small datasets introduces unique challenges such as overfitting (Chen et al., 2021), bias (Luo and Paal, 2021), model instability, difficulties in model selection but also general lack of performance (Maimon and Rokach, 2007). Therefore it is even more important to carefully find and hypertune an appropriate model, while acknowledging its strengths and weaknesses.

Although limited data availability is still a common issue in many fields, progress in machine learning for small datasets has been slower compared to large-scale learning, largely due to the difficulty of the task (Dou et al., 2023). This chapter aims not only to identify the most effective models for our specific problem but also to contribute to the broader discussion on the reliability of supervised learning in data-scarce environments. Fields like healthcare, molecular chemistry and material sciences are known for researching

on methods to tackle small dataset challenges (Pais et al., 2024; Achar and Keith, 2024). Two promising approaches (Wen et al., 2024) which we will cover use a virtual sample generation process in order to make the training dataset bigger.

4.2 Virtual Sample Generation (VSG)

Virtual Sample Generation is a relatively new method that aims at improving the performance of data-based modeling by generating additional virtual samples. Virtual samples refer to auxiliary samples, which are generated using prior knowledge and the available training data. It can close information gaps caused by insufficient real data, increasing the number of samples available for fitting a model. This may result in improving the predictive ability of the model and counteract overfitting (Li et al., 2021). In materials science, where gathering data is both difficult and expensive, VSG methods have become a popular workaround and are still being further refined (Yang et al., 2011; Xu et al., 2023; Chen et al., 2023). We will apply two of these VSG techniques, since their ability to enlarge small datasets for improved model fitting will be very useful in the case of our small data set.

Notation: In this section, let $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{n} \subset \mathbb{R}^{p+1}$ denote a dataset consisting of n observations, where each $\mathbf{x}_i \in \mathbf{X}$ consists of p predictor variables together with the corresponding response variable.

4.2.1 Gaussian Mixture Model VSG

In this approach, virtual samples are generated using a method called the GMM-VSG method (Shen and Qian, 2022). It generates virtual data by fitting a Gaussian Mixture Model (GMM) to the available training data and then creates virtual samples from that estimated distribution. By adding the virtual samples to the training data, models might be able to better understand the training data structure, improving its predictive ability without overfitting to the data. Shen and Qian (2022) have found it particularly effective for small, clustered datasets especially within the topic of material sciences.

Gaussian Mixture Model

A GMM is a parametric probability density function represented as a weighted sum of Gaussian component densities (Bera and Bilias, 2002). We will estimate the GMM parameters using the iterative Expectation-Maximization (EM) algorithm.

A Gaussian mixture model of some dataset $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{n}$ is a weighted sum of M component Gaussian densities given by

$$p(\mathbf{x}_i \mid \lambda) = \sum_{j=1}^{M} w_j \, g(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \Sigma_j), \quad \lambda = \{\lambda_j = \{w_j, \boldsymbol{\mu}_j, \Sigma_j\}\}_{j=1}^{M}$$

where:

- $w_j,\ j = 1, \ldots, M$ are the mixture weights, which represent the weight given to component j, satisfying $\sum_{j=1}^{M} w_j = 1$,
- $g(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \Sigma_j), j = 1, \ldots, M$ are the component Gaussian densities,

- every component density is a d-variate Gaussian function of the form

$$g(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \Sigma_j) = \frac{1}{(2\pi)^{d/2}|\Sigma_j|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu}_j)^\top \Sigma_j^{-1}(\mathbf{x}_i - \boldsymbol{\mu}_j)\right),$$

  with mean vector $\boldsymbol{\mu}_j$ and covariance matrix Σ_j.

The complete Gaussian mixture model is parameterized by the mean vectors, covariance matrices and mixture weights from all component densities. Given training data and a GMM configuration, λ_j can be estimated to approximately match the distribution of the training data.

There are several techniques available for estimating the parameters of a GMM (Reynolds et al., 2009). The most popular and well-established method is the maximum likelihood (ML) estimation. The aim of ML estimation is to find the model parameters which maximize the likelihood of the GMM given the training data. For $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, the GMM likelihood can be written as

$$p(\mathbf{X}|\lambda) = \prod_{i=1}^{n} p(\mathbf{x}_i|\lambda).$$

Unfortunately, this expression is a non-linear function of the parameters λ and direct maximization is not possible. However, ML parameter estimates can be obtained iteratively using a special case of the expectation-maximization (EM) algorithm (McLachlan and Basford, 1988).

EM Algorithm

The basic concept of the EM algorithm is, beginning with an initial model λ, to estimate a new model $\tilde{\lambda}$, such that $p(\mathbf{X}|\tilde{\lambda}) \geq p(\mathbf{X}|\lambda)$. On each EM iteration, the following re-estimation formulas are used which guarantee a monotonic increase in the model's likelihood value.

i.) **E-Step**: Compute the probability for every component j. The *a posteriori* probability for component j is given by

$$\Pr(j|\mathbf{x}_i, \lambda) = \frac{w_j g(\mathbf{x}_i|\boldsymbol{\mu}_j, \Sigma_j)}{\sum_{k=1}^{M} w_k g(\mathbf{x}_i|\boldsymbol{\mu}_k, \Sigma_k)}.$$

ii.) **M-Step**: Update estimates for every component j.

- Mixture Weights: $\tilde{w}_j = \frac{1}{n}\sum_{i=1}^{n} \Pr(j|\mathbf{x}_i, \lambda)$,
- Means: $\tilde{\boldsymbol{\mu}}_j = \frac{\sum_{i=1}^{n} \Pr(j|\mathbf{x}_i,\lambda)\mathbf{x}_i}{\sum_{i=1}^{n} \Pr(j|\mathbf{x}_i,\lambda)}$,
- Variances (Diagonal Covariances): $\tilde{\sigma}_j^2 = \frac{\sum_{i=1}^{n} \Pr(j|\mathbf{x}_i,\lambda)x_i^2}{\sum_{i=1}^{n} \Pr(j|\mathbf{x}_i,\lambda)} - \tilde{\mu}_j^2$,

  where σ_j^2, x_i and μ_j refer to arbitrary elements of the vectors $\boldsymbol{\sigma}_j^2$, $\mathbf{x}_i$ and $\boldsymbol{\mu}_j$.

These two steps are repeated until some convergence threshold is reached. More on the EM algorithm can be found in Moon (1996).

Since the solution of the EM algorithm can strongly depend on the initial model chosen, it is crucial to properly initialize it. For this, we will use the search / run / select (S-R-S) strategy from Biernacki et al. (2003):

i.) **Search**: For each number of components M on the grid $\mathcal{M} = \{1, \ldots, M_{\max}\}$, p initial positions are created; these can be based on either a classification EM (CEM) algorithm, a stochastic EM (SEM) algorithm or short runs of the standard EM algorithm.

ii.) **Run**: The EM algorithm is run a set number of times at each initial position with a fixed number of iterations.

iii.) **Select**: The solution with the best likelihood is selected.

Handling categorical predictors

For categorical predictors, we follow the Gaussian × Multinomial approach first laid out by Lawrence and Krzanowski (1996).

We split every observation into

$$\mathbf{x}_i = \left(\mathbf{z}_i^\top, \mathbf{y}_i^\top\right)^\top, \qquad \mathbf{z}_i \in \mathbb{R}^p, \quad \mathbf{y}_i = (y_{i1}, \ldots, y_{iq}),\ y_{ir} \in \{1, \ldots, L_r\},$$

where $\mathbf{z}_i$ denote the continuous variables and $\mathbf{y}_i$ denote the categorical variables of $\mathbf{x}_i$.

Every mixture component j ($j = 1, \ldots, M$) factorizes into a Gaussian block for the continuous part and a product of multinomials for the categorical part,

$$f_j(\mathbf{x}_i \mid \lambda_j) \;=\; g(\mathbf{z}_i \mid \boldsymbol{\mu}_j, \Sigma_j) \prod_{r=1}^{q} \mathrm{Mult}(y_{ir} \mid \boldsymbol{\phi}_{jr}), \qquad \sum_{\ell=1}^{L_r} \phi_{jr\ell} = 1.$$

Here, $\boldsymbol{\phi}_{jr} = (\phi_{jr1}, \ldots, \phi_{jrL_r})$ collects the level probabilities of factor r inside component j. The global mixture is therefore given by

$$p(\mathbf{x}_i \mid \lambda) = \sum_{j=1}^{M} w_j\, f_j(\mathbf{x}_i \mid \lambda_j), \qquad \lambda_j = \{\boldsymbol{\mu}_j, \Sigma_j, \boldsymbol{\phi}_{j1}, \ldots, \boldsymbol{\phi}_{jq}\}.$$

Gaussian Mixture Model VSG

We now take a closer look at how virtual samples are generated using the Gaussian Mixture Model approach.

Assume we have a training dataset $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{R}^p$, where we have p predictor variables. In this case we denote the response for each $\mathbf{x_i}$ by y_i.

***Toy example**: Consider the following simple linear model*

$$y = 1 + 2x + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, 1).$$

We generate 1000 datasets based on this distribution, each consisting of 7 *observations generated with different random noise ε. On each of the 1000 datasets, we will run the oversampling method independently to evaluate the performance of the method in the linear setting. We summarize the results in Table 4.2.*

One such dataset, which is given in Table 4.1, will be used throughout the method in order to illustrate it.

x	y
1	3.75
2	7.01
3	6.64
4	11.03
5	8.78
6	13.76
7	13.69

Table 4.1: Simulated observations of $y = 1 + 2x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1)$.

We proceed as follows:

i.) **S-R-S**: For every candidate component count $M \in \mathcal{M} = \{1, \ldots, M_{\max}\}$, the S-R-S strategy of Biernacki et al. (2003) provides the mixture parameters set

$$\widehat{\lambda}^{(M)} = \{\, \widehat{w}_j, \widehat{\boldsymbol{\mu}}_j, \widehat{\Sigma}_j, \widehat{\boldsymbol{\phi}}_{j1}, \ldots, \widehat{\boldsymbol{\phi}}_{jq} \}_{j=1}^{M}$$

together with its maximized log-likelihood $\log \mathcal{L}\big(\widehat{\lambda}^{(M)}\big)$.

***Toy example**: Fit mixtures with $M_{\max} = 7$ on the 7 observations of training dataset.*

ii.) **Component Selection:** By using the Akaike Information Criterion (AIC), we determine the best component count:

$$\widehat{M} \;=\; \arg\min_{M \in \mathcal{M}} \mathrm{AIC}(M) \;=\; \arg\min_{M} \Big[-2 \log \mathcal{L}\big(\widehat{\lambda}^{(M)}\big) + 2 \,\big\|\widehat{\lambda}^{(M)}\big\| \Big].$$

Let $\widehat{\lambda} = \widehat{\lambda}^{(\widehat{M})}$ for the remainder of the procedure.

iii.) **Sampling**: Sample from the selected mixed Gaussian distribution.

(a) Draw a component index $K \sim \mathrm{Cat}\big(\widehat{w}_1, \ldots, \widehat{w}_{\widehat{M}}\big)$.

(b) Generate one synthetic feature vector $\mathbf{x}_{\text{virtual}} = (\mathbf{z}_{\text{virtual}}^{\top}, \mathbf{t}_{\text{virtual}}^{\top})^{\top}$ conditional on K:

$$\mathbf{z}_{\text{virtual}} = \widehat{\boldsymbol{\mu}}_K + \widehat{L}_K \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, I_p), \quad \widehat{L}_K \widehat{L}_K^{\top} = \widehat{\Sigma}_K,$$

$$\mathbf{t}_r^{\text{virtual}} \sim \mathrm{Cat}\big(\widehat{\boldsymbol{\phi}}_{Kr}\big), \qquad r = 1, \ldots, q.$$

(c) Using the unweighted k nearest neighbors of $\mathbf{x}_{\text{virtual}}$ in $\mathbf{X}$, assign a pseudo-response $y_{\text{virtual}} = \frac{1}{k} \sum_{i \in \mathcal{N}_k(\mathbf{x}_{\text{virtual}})} y_i$.

***Toy example**: Draw $m = 40$ virtual x values; for each, compute $y_{virtual}$ with $k = 3$.*

iv.) **Domain restriction**: For variables where the range of the feature space is known, we discard the virtual samples which have values outside of the range.

***Toy example**: To conclude our example, we fit two linear regressions: one on the final dataset ($n = 45$) obtained above, and one on the training observations ($n = 7$). For each of the 1000 datasets, we now randomly draw 4 points from the interval $[1, 7]$, compute their corresponding response $y = 1 + 2x$ and predict their outcome with the two models. To compare the two models, we look at the mean RMSE of the 1000 datasets with and without virtual samples. We also look at the mean bias and variance of the estimator of*

the model. They are reported in Table 4.2. We note that there is no significant difference in the group means of either metric given a paired t-test.

	With VSG	Without VSG	Means Diff. 95% CI
Mean RMSE	0.532	0.440	$[0.076, 0.107]$
Mean Bias	0.012	0.014	$[-0.017, 0.013]$
Mean Variance	0.150	0.083	$[0.056, 0.080]$

Table 4.2: Mean RMSE, bias and variance and the corresponding 95% confidence interval of the paired t-test of the linear models with and without virtual samples generated with the GMM-VSG method over the 1000 runs.

4.2.2 Adapted Bush Topology SMOTE

Virtual Sample Generation with oversampling methods is common in the classification setting to overcome the problem of unbalanced data (Elreedy and Atiya, 2019; Tao et al., 2020). However, little attention has been paid to oversampling methods for regression problems. One method that is used for oversampling is called Synthetic Minority Oversampling TEchnique (SMOTE) (Chawla et al., 2002). It is designed to address class imbalance by creating synthetic samples of the minority class. Rather than simply duplicating existing samples, SMOTE generates new data points by interpolating between a minority sample and one of its k nearest minority neighbors in the feature space. The classic SMOTE method works as follows:

i.) **Identify the Minority Class:** Let the set of minority class samples be denoted as:

$$\mathcal{X}_{\text{min}} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}, \quad \mathbf{x}_i \in \mathbb{R}^{p+1}$$

ii.) **Choose Number of Nearest Neighbors:** For each sample $\mathbf{x}_i$, identify its k-nearest neighbors within the minority class (the default is $k = 5$), using Euclidean distance:

$$\text{NN}_k(\mathbf{x}_i) = \{\mathbf{x}_{i_1}, \ldots, \mathbf{x}_{i_k}\}$$

iii.) **Determine Number of Synthetic Samples:** Decide on a sampling ratio (e.g., 200%, which means generating two synthetic samples for each real sample). Based on this, calculate the number of new synthetic samples to generate for each $\mathbf{x}_i$ by multiplying $n \cdot \frac{\text{sampling ratio}}{100}$.

iv.) **Generate Synthetic Samples:**

(a) For each sample, randomly select one of the k nearest neighbors, $\mathbf{x}_{\text{nn}}$.

(b) Create the synthetic sample by linear interpolation:

$$\mathbf{x}_{\text{virtual}} = \mathbf{x}_i + \lambda \cdot (\mathbf{x}_{\text{nn}} - \mathbf{x}_i), \quad \lambda \sim \mathcal{U}(0, 1)$$

v.) **Add Synthetic Samples to Dataset:** Append the generated synthetic samples $\mathbf{x}_{\text{virtual}}$ to the training dataset. These new samples belong to the minority class.

Some remarks:

- SMOTE assumes a continuous/numeric feature space. For categorical features, adaptations like SMOTE-NC can be used.

- Synthetic points lie along the line segments between real minority class instances; no extrapolation occurs.
- SMOTE can lead to class overlap if the minority class is not well separated from the majority.

The challenge of using SMOTE for regression lies in determining the region of a feature space to be over-sampled. For classification problems, this is easily done by determining the minority class. However, for regression problems, an estimate function is required to represent some target value in a certain feature space (Pelayo and Dick, 2019). In Sutojo et al. (2020) oversampling in regression is done by using the SMOTE and distributing synthetic data according to Bus, Star, and Mesh topologies which are defined below. This overcomes the issue of determining the region of a feature space.

- Bus Topology

  Order the samples (e.g. by the first principal-component score or any meaningful index) and connect each sample to its immediate neighbor, mirroring stops along a bus route. For every consecutive pair $(\mathbf{x}_i, \mathbf{x}_{i+1})$ create

  $$\mathbf{x}_{\text{virtual}} = \mathbf{x}_i + \lambda(\mathbf{x}_{i+1} - \mathbf{x}_i), \qquad \lambda \sim \mathcal{U}(0,1).$$

  This places synthetic samples evenly along the path that traverses the data in sequence.

- Star Topology

  Let the samples be $\{\mathbf{x}_i\}_{i=1}^n$ and their centroid $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i$. Each sample is connected to the centroid, forming a set of radial "spokes". For every pair $(\mathbf{x}_i, \bar{\mathbf{x}})$ generate

  $$\mathbf{x}_{\text{virtual}} = \mathbf{x}_i + \lambda(\bar{\mathbf{x}} - \mathbf{x}_i), \qquad \lambda \sim \mathcal{U}(0,1),$$

  so the synthetic points radiate around the center of the minority cloud.

- Mesh Topology

  Form a complete graph by connecting every sample with every other sample; $\binom{n}{2}$ pairs in total. For every pair of samples $(\mathbf{x}_p, \mathbf{x}_q)$ generate a dense "mesh":

  $$\mathbf{x}_{\text{virtual}} = \mathbf{x}_p + \lambda(\mathbf{x}_q - \mathbf{x}_p), \qquad \lambda \sim \mathcal{U}(0,1), \quad \forall\, p < q.$$

Bush Topology SMOTE

In Sutojo et al. (2023), a revisited, more flexible approach called Bush Topology is proposed, building on the initial Bus Topology defined above. By comparing several Hamiltonian paths, the Bush Topology preserves the interpolation of the bus topology SMOTE while selectively filling gaps that most benefit the predictive model, leading to further gains in RMSE on the same small-data benchmarks according to Sutojo et al. (2023).

Note: A Hamiltonian path is a path in a graph that visits every vertex exactly once. In our case, a Hamiltonian path is a path which visits every sample of our dataset exactly once.

Assume we have a training dataset $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{R}^{p+1}$ consisting of n training samples each containing predictors and a response.

***Toy example**: Consider the following simple linear model*

$$y = 1 + 2x + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, 1).$$

We generate 1000 datasets based on this distribution, each consisting of 7 observations generated with different random noise ε. On each of the 1000 datasets, we will run the oversampling method independently to evaluate the performance of the method in the linear setting. We summarize the results in Table 4.3.

One such dataset, which is given in Table 4.1, will be used throughout the oversampling method in order to illustrate it.

We proceed as follows:

i.) **Monte Carlo Cross Validation:** Split the training dataset randomly into training and validation sets according to the Monte Carlo Cross Validation (MCCV) method (Xu and Liang, 2001).

***Toy example**: Draw 5 points for training; $x = 1, 3, 4, 5, 7$, leaving $x = 2, 6$ for validation.*

ii.) **Compute Virtual Samples:** Assume the obtained training set contains ℓ samples. Ignoring the start-point direction, there are $\frac{(\ell-1)!}{2}$ distinct Hamiltonian paths. A *Hamiltonian path* is an ordering $(\mathbf{x}_{\pi(1)}, \ldots, \mathbf{x}_{\pi(\ell)})$ that visits every sample once. For each path, we proceed as follows:

(a) **Interpolation:** For each consecutive pair $(\mathbf{x}_i, \mathbf{x}_{i+1})$, $i = 1, \ldots, \ell - 1$ on the path, generate m virtual points using

$$\mathbf{x}_{\text{virtual}} = \mathbf{x}_i + t(\mathbf{x}_{i+1} - \mathbf{x}_i), \quad t \in \left\{\tfrac{1}{m+1}, \ldots, \tfrac{m}{m+1}\right\}.$$

This yields $(\ell - 1) \cdot m$ virtual samples per path (we set $m = 10$).

***Toy example**: We have a natural ordering $(1, 3, 4, 5, 7)$. Let $m = 10$; we generate m equidistant points for every pair. E.g. for the pair $(1, 3.75)$ and $(3, 6.64)$ we get* (1.18, 4.02), (1.36, 4.38), (1.55, 4.54), (1.73, 4.80), (1.91, 5.07), (2.09, 5.33), (2.27, 5.59), (2.45, 5.86), (2.64, 6.12), (2.82, 6.38).

(b) **Temporary Model Training:** Train a temporary model on the $(\ell - 1) \cdot m$ virtual and the ℓ actual training samples.

***Toy example**: We fit a temporary linear regression $y \sim x$ with our $4 \cdot 10$ virtual and 5 actual training samples.*

(c) **Temporary Model Validation:** Validate the temporary model on the validation set obtained in step i. Compute its RMSE.

***Toy example**: We validate the linear regression on the validation observations $x = 2, 6$. We obtain a RMSE of* 1.87.

iii.) **Path selection:** Choose the k paths with the lowest RMSE and compute their average RMSE (the paper sets $k = 10$).

iv.) **Choose final virtual samples:** If the average RMSE is lower than the average RMSE from the previous cycle, replace the saved virtual samples with the virtual samples from the current k paths chosen in step iii.

v.) **Repeat Step i to iv:** Repeat steps i to iv until the average RMSE doesn't further decrease for three iterations. Then we can combine the saved virtual samples with the ℓ training samples, giving a total of $(\ell - 1) \cdot m \cdot k + \ell$ samples.

***Toy example**: We change the training / validation split of the observations given in Table 4.1 and compute the virtual samples once again. We refit a temporary model and report RMSE on the validation set. We repeat this until the RMSE doesn't decrease for three iterations.*

vi.) **Train Final Model:** Train the final model with the dataset constructed in step v. Use this model as needed.

***Toy example**: To conclude our example, we fit two linear regressions: one on the final dataset ($n = 45$) obtained above, and one on the final training observations ($n = 5$). For each of the 1000 datasets, we now randomly draw 4 points from the interval $[1, 7]$, compute their corresponding response $y = 1 + 2x$ and predict their outcome with the two models. To compare the two models, we look at the mean RMSE of the 1000 datasets with and without virtual samples. We also look at the mean bias and variance of the estimator of the model. They are reported in Table 4.3. We note that there is no significant difference in the group means of either metric given a paired t-test.*

	With VSG	Without VSG	Means Diff. 95% CI
Mean RMSE	0.480	0.470	$[-0.001, 0.020]$
Mean Bias	0.006	0.002	$[-0.007, 0.015]$
Mean Variance	0.11	0.10	$[-0.003, 0.012]$

Table 4.3: Mean RMSE, bias and variance and the corresponding 95% confidence interval of the paired t-test of the linear models with and without virtual samples generated with the Adapted Bush Topology SMOTE over the 1000 runs.

Handling categorical predictors (SMOTE-NC)

The classic SMOTE applies only to numeric attributes. For categorical variables it needs an adapted version, which is given by the Synthetic Minority Over-sampling TEchnique - Nominal Continuous (SMOTE-NC) (Chawla et al., 2002).

For the Bush SMOTE, the synthetic values $\mathbf{x}^{(j)}_{\text{virtual}}$ for every categorical feature j are generated as follows:

$$\mathbf{x}^{(j)}_{\text{virtual}} = \begin{cases} \mathbf{x}^{(j)}_i, & \mathbf{x}^{(j)}_i = \mathbf{x}^{(j)}_{i+1}, \\ \text{rand}\{\mathbf{x}^{(j)}_i, \mathbf{x}^{(j)}_{i+1}\}, & \mathbf{x}^{(j)}_i \neq \mathbf{x}^{(j)}_{i+1}, \end{cases}$$

where $\mathbf{x}_i$ and $\mathbf{x}_{i+1}$ are a consecutive pair on a Hamiltonian path in the Adapted Bush Topology SMOTE.

4.3 Experimental Setup

To obtain a fair comparison of every supervised-learning model, we define a universal experimental pipeline. Each model, irrespective of its internal mechanics, undergoes the

same train–validation protocol and is finally evaluated on the 2024 Swiss Climbing Team dataset.

i.) **Training dataset:** The model is trained on one of the three datasets detailed below. These datasets differ in the way virtual samples are generated and added; all actual samples are from the 2023 training dataset.

ii.) **Hyper-parameter tuning:** Each dataset is processed with a five-fold cross validation loop: we refit the model on every training folds, and choose its hyper-parameters to minimize the validation error across the folds. This ensures that the resulting model delivers strong predictive accuracy without overfitting.

iii.) **Model evaluation:** The tuned model is used to predict the Performance Score for the 2024 Swiss Climbing Team dataset which was extensively explored in the EDA chapter. The accuracy is reported with the root mean squared error (RMSE) and R squared ($\boldsymbol{R}^2$).

Note: The coefficient of determination, also called R squared ($\boldsymbol{R}^2$) is defined as follows:

$$\boldsymbol{R}^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Training datasets

- **2023 Swiss Climbing Team dataset:** The base dataset contains the physiological screening results of the 2023 national climbing team. The 15 athletes of the team are measured on 13 variables, plus the composite Performance Score defined in Chapter 2.
- **Base Data + Gaussian-Mixture VSG:** Following the protocol of Shen and Qian (2022), we fit a Gaussian mixture model to the original 2023 data and draw 1000 synthetic samples. The resulting set contains 1015 rows. Performance Scores of virtual points are truncated to the interval $[0, 4000]$, reflecting the possible lower and upper bounds of the metric by definition. We further set the maximum number of mixture components to $M_{\max} = 14$, use $p = 20$ random initial positions for each component count, and draw the response from the $k = 5$ nearest neighbors.
- **Base Data + Bush-SMOTE VSG:** Here we interpolate along Hamiltonian paths through the 15 real points, as proposed by Sutojo et al. (2023). We adopt their recommended hyper-parameters $m = 10$ interpolation points per segment, $k = 10$ best paths retained per cross validation split, and the suggested 70/30 Monte-Carlo cross validation split. For evaluating the best paths per split, we use a random forest validation model with num.trees = 200, since this yields lower cross validation error than the kNN algorithm suggested in the paper. We furthermore cap the selection of Hamiltonian paths at 50 paths per split. The exhaustive enumeration would explode to approximately $2 \cdot 10^5$ paths, which is computationally infeasible. Duplicate synthetic rows are removed before modeling since it produced lower error. This results in 788 unique virtual samples, for a total of 803 observations.

4.4 Comparison of Supervised Learning Models

In order to compare the different supervised learning models, we look at the root mean squared error (RMSE) and the coefficient of determination ($\boldsymbol{R}^2$). In Figure 4.1, we summa-

rize the RMSE of the different supervised learning models for the three different training sets. The details will be given in subsections 4.4.1 - 4.4.10.

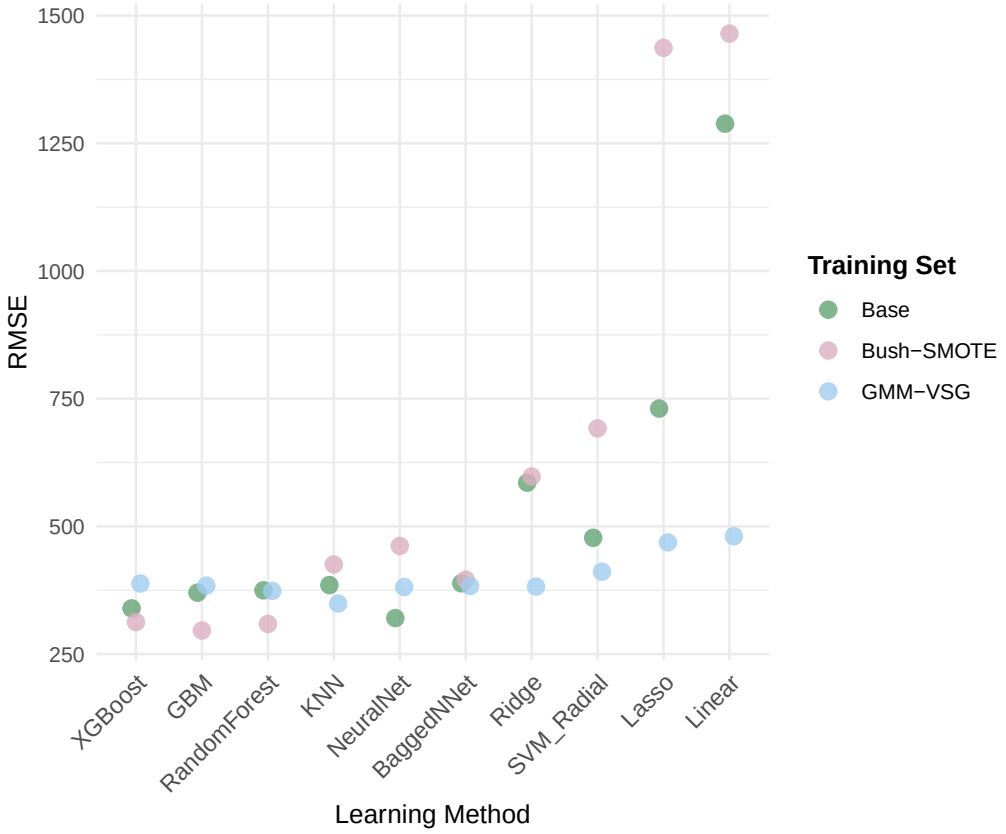


Figure 4.1: Comparison of the RMSE of the different supervised learning models given the dataset used for training.

Table 4.4 shows that tree-based models trained on the Bush-SMOTE dataset give the lowest RMSE and the highest $\boldsymbol{R}^2$. Compared with the base training dataset, Bush-SMOTE improved the performance as follows:

- **Gradient Boosting Machine:** RMSE improves from 370.2 to 296.2 (–20.0%); R^2 increases from 0.175 to 0.576 (+0.401).
- **Random Forest:** RMSE improves from 374.8 to 309.1 (–17.5%); R^2 increases from 0.199 to 0.492 (+0.293).
- **XGBoost:** RMSE improves from 339.5 to 312.7 (–7.9%); R^2 increases from 0.346 to 0.624 (+0.278).

For all distance-based models (kNN, Ridge, SVM, Lasso and OLS), GMM-VSG outperforms the other two training datasets. The base dataset without additional virtual samples only produces the best RMSE when fitted to the simple neural network. In the case of bagged neural networks, all three datasets give similar results.

Overall, the tree-based models produced the lowest RMSE and the highest $\boldsymbol{R}^2$, confirming that the models are robust even with non-linear, heterogeneous data structures and heteroscedastic noise.

Training Set	Model	RMSE	R^2
Base	BaggedNNet	388.6	0.104
	GBM	370.2	0.175
	KNN	385.1	0.502
	Lasso	730.6	0.005
	Linear	1288.4	0.032
	NeuralNet	320.5	0.470
	RandomForest	374.8	0.199
	Ridge	585.6	0.001
	SVM_Radial	477.7	0.399
	XGBoost	339.5	0.346
Bush–SMOTE	BaggedNNet	395.7	0.471
	GBM	**296.2**	0.576
	KNN	425.5	0.260
	Lasso	1436.9	0.082
	Linear	1464.8	0.082
	NeuralNet	461.6	0.485
	RandomForest	309.1	0.493
	Ridge	597.6	0.036
	SVM_Radial	691.8	0.007
	XGBoost	312.7	**0.624**
GMM–VSG	BaggedNNet	383.2	0.123
	GBM	384.0	0.123
	KNN	348.9	0.267
	Lasso	468.6	0.001
	Linear	480.7	0.003
	NeuralNet	381.3	0.139
	RandomForest	373.5	0.199
	Ridge	382.1	0.130
	SVM_Radial	411.3	0.030
	XGBoost	388.1	0.111

Table 4.4: RMSE and $\boldsymbol{R}^2$ of every supervised learner evaluated on the 2024 Swiss Climbing Team dataset for the three training datasets. Bold numbers mark the lowest RMSE and highest $\boldsymbol{R}^2$.

4.4.1 Linear Regression

The linear regression, also known as ordinary least squares method minimizes the empirical squared loss between the observed outcome and the output of the linear function, given by

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2.$$

Under the Gauss-Markov assumptions of exogenity, full rank and homoscedasticy of the errors, OLS produces the best linear unbiased estimator (BLUE), meaning lowest variance for an unbiased estimator (Chipman, 2011). Despite its simplicity, OLS remains a popular method. Firstly, because inference on individual predictors is straightforward via t– and F-statistics, and secondly, since the fitted values form the basis for many more complex models (e.g. boosting).

Fitted on the **Bush-SMOTE** training set, OLS yields RMSE = 1460, $\boldsymbol{R}^2 = 0.082$, suggesting that the relationship of our variables is not linear.

4.4.2 Ridge Regression

One downside of OLS is that given multicollinearity, the variance of the estimates can inflate; ridge regression counteracts this by augmenting the loss with an ℓ_2 penalty,

$$\hat{\boldsymbol{\beta}}(\lambda) = \arg\min_{\boldsymbol{\beta}} \Big\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \Big\},$$

where the regularization path trades bias for variance reduction (Hoerl, 2020). Because the penalty shrinks all coefficients simultaneously it can improve numerical stability and prediction error when the features are highly correlated. We tuned the model by testing a log-grid with 50 values for λ through inner cross validation.

Fitted on the **GMM-VSG** training set, ridge regression yields RMSE= 382, $\boldsymbol{R}^2 = 0.130$, which shows that we successfully counteracted the multicollinearity.

4.4.3 Lasso Regression

The least-absolute-shrinkage-and-selection operator (Lasso) is similar to the ridge regression, but replaces the ℓ_2 norm by an ℓ_1 norm,

$$\hat{\boldsymbol{\beta}}(\lambda) = \arg\min_{\boldsymbol{\beta}} \Big\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \Big\}.$$

This norm sets less informative coefficients exactly to zero, in other words performing feature selection in addition to regularization. This is especially useful in high-dimensional settings, where we have many features but limited samples. However, lasso regression assumes that the coefficients of the linear model are sparse. If not the case, the model can induce a big bias (Ranstam and Cook, 2018). We tuned the model by using coordinate descent over 100 different λ values.

Fitted on the **Bush-SMOTE** training set, lasso regression yields RMSE = 1440, $\boldsymbol{R}^2 = 0.082$, indicating that the underlying structure is not well represented by a sparse linear model.

4.4.4 k-Nearest-Neighbors Regression

The k-nearest-neighbors algorithm is a non-parametric supervised learning model. For a new sample $\mathbf{x} = (x_1, \ldots, x_n)$ with unknown outcome $\mathbf{y}$, the prediction is given by

$$\hat{f}_k(\mathbf{x}) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(\mathbf{x})} y_i,$$

where $\mathcal{N}_k(\mathbf{x})$ is the set of the k closest observations under some chosen metric. In local neighborhoods of the training observations, the new sample $\mathbf{x}$ is expected to have a similar outcome $\hat{f}_k(\mathbf{x})$ like the outcomes of the training samples. Hence, for an unknown $\mathbf{y}$, the value must be similar to the values of the closest observations, which is modeled by the average of the k-nearest observations Kramer (2013). We tuned the model by centering and scaling the inputs, limited the number of neighbors k to at most 15, examined both Euclidean ($p = 2$) and Manhattan ($p = 1$) distances and tried various kernels.

Fitted on the **Base** training set, the tuned kNN algorithm yields RMSE = 385, $\boldsymbol{R}^2 = 0.502$.

4.4.5 Support Vector Regression with RBF Kernel

Classic support–vector machines (SVMs) seek the maximum margin hyperplane that separates two classes, introducing slack variables when the classes are not linearly separable (Cortes and Vapnik, 1995). The separating hyperplane is determined uniquely by a subset of training points called support vectors. The sparse solution and good generalization of the support vector machine lend themselves to adaptation to regression problems. By replacing the binary loss with an ε–insensitive loss called ε-tube, one obtains support vector regression (SVR), which fits a "tube" of radius ε around the unknown regression function and penalizes only errors outside that tube (Awad and Khanna, 2015).

With training inputs $\mathbf{x}_i \in \mathbb{R}^p$ and responses $y_i \in \mathbb{R}$ the SVR optimization is given by

$$\min_{\mathbf{w},b,\xi,\xi^*} \tfrac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \quad \text{s.t.} \quad \begin{cases} y_i - \mathbf{w}^\mathsf{T}\phi(\mathbf{x}_i) - b \le \varepsilon + \xi_i, \\ \mathbf{w}^\mathsf{T}\phi(\mathbf{x}_i) + b - y_i \le \varepsilon + \xi_i^*, \\ \xi_i,\ \xi_i^* \ge 0. \end{cases},$$

where:

- $\mathbf{x}_i \in \mathbb{R}^p$ is the input vector of the i-th sample,
- $y_i \in \mathbb{R}$ is the corresponding response,
- $\phi(\cdot)$ is a nonlinear feature mapping into a high-dimensional Hilbert space,
- $\mathbf{w}$ is a weight vector that defines the hyperplane,
- b is the bias (offset) of the hyperplane,
- $\varepsilon > 0$ is the radius of the ε-tube,
- $\xi_i, \xi_i^* \ge 0$ are slack variables for points lying above or below the tube,
- $C > 0$ is a regularization constant trading off flatness ($\|\mathbf{w}\|_2^2$) and tube violations.

In the dual, the kernel trick replaces inner products by $K(\mathbf{x}, \mathbf{z})$. The fitted support vector regression predicts a new input as

$$\hat{\mathbf{y}} = \sum_{i \in \mathcal{S}} (\alpha_i - \alpha_i^*) K(\mathbf{x_i}, \mathbf{x}_{\text{new}}) + b,$$

where $\mathcal{S}$ is the support-vector index set and α_i, α_i^* are optimal dual coefficients. We adopted an radial basis function (RBF) kernel given by $K_{\text{RBF}}(\mathbf{x}, \mathbf{z}) = \exp\!\left(-\frac{\|\mathbf{x}-\mathbf{z}\|_2^2}{2\sigma^2}\right)$. We tuned the model by centering and scaling the inputs and initialized the kernel bandwidth σ by the median heuristic $\sigma_{\text{median}} = \text{median}\{\|\mathbf{x}_i - \mathbf{x}_j\|_2 : 1 \le i < j \le n\}$ for $C \in \{0.5, 1, 2, 4\}$.

Fitted on the **Base** training set, the tuned support vector regression yields RMSE = 478, $\boldsymbol{R}^2 = 0.399$.

4.4.6 Random Forests

Random forests are an ensemble model that aggregates $B \in \mathbb{N}$ regression trees grown on bootstrap samples (Breiman, 2001). Let $\mathbf{x}_i \in \mathbb{R}^p$ and $y_i \in \mathbb{R}$, $\quad i = 1, \ldots, n$ denote the training data, and let $h(\mathbf{x}; \Theta_b)$ be the predictor of the b-th tree, where the Θ_b are i.i.d.

random variables. The random forest predictor is given by

$$\hat{f}_{\mathrm{RF}}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} h(\mathbf{x};\Theta_b), \qquad \mathbf{x}\in\mathbb{R}^p.$$

Following Breiman (2001), every tree in the ensemble is grown with two independent sources of randomness

$$\Theta_b = (\mathcal{I}_b, \mathcal{F}_b), \quad b = 1,\dots,B,$$

where:

- $\mathcal{I}_b$ is a bootstrap sample of size n drawn with replacement from $\{1,\dots,n\}$,
- $\mathcal{F}_b$ is the sequence of feature–subsets chosen at each split.

We write the partition induced by the b-th tree as $\mathcal{P}_b = \{R_{b_1},\dots,R_{b_M}\}$ with terminal regions $R_{b_M}\subset\mathbb{R}^p$. The tree predictor can be expressed as the step function

$$h(\mathbf{x};\Theta_b) = \sum_{m=1}^{M} \bar{y}_{b_M}\,\mathbb{1}_{\{\mathbf{x}\in R_{b_M}\}}, \qquad \bar{y}_{b_M} = \frac{1}{|\mathcal{I}_b\cap R_{b_M}|}\sum_{i\in\mathcal{I}_b\cap R_{b_M}} y_i,$$

which results in $\hat{f}_{\mathrm{RF}}(\mathbf{x}) = \mathbb{E}_\Theta[h(\mathbf{x};\Theta)]$ for $B\to\infty$ by the law of large numbers. We tuned the model by setting $B = 500$ and by performing grid search over the number of randomly chosen variables used for determining the splits $m\in\{1,4,7,10,14\}$ and the minimum node sizes 1, 5 and 10.

Fitted on the **Bush-SMOTE** training set, the tuned random forest yields RMSE = 309, $\boldsymbol{R}^2 = 0.492$.

4.4.7 Gradient Boosting Machine (GBM)

Gradient boosting can be interpreted as functional gradient descent in the space of additive functions. At iteration m, we seek a weak learner $f_m(\mathbf{x})$ that most steeply reduces the empirical risk

$$\mathcal{L}(F) \;=\; \sum_{i=1}^{n}\ell\big(y_i, F(\mathbf{x}_i)\big),$$

with a differentiable loss ℓ such as the squared error, absolute error or the negative binomial log-likelihood (Friedman, 2001). Given the current model F_{m-1}, we compute pseudo-residuals

$$r_{i_m} = -\frac{\partial\ell\big(y_i, F_{m-1}(\mathbf{x}_i)\big)}{\partial F_{m-1}(\mathbf{x}_i)}, \quad i = 1,\dots,n,$$

fit a depth-d regression tree f_m to the pairs $\{(\mathbf{x}_i, r_{i_m})\}$, determine region-wise line-search solutions

$$\gamma_{j_m} \;=\; \arg\min_{\gamma}\sum_{\mathbf{x}_i\in R_{jm}}\ell\big(y_i, F_{m-1}(\mathbf{x}_i)+\gamma\big),$$

and update

$$F_m(\mathbf{x}) \;=\; F_{m-1}(\mathbf{x}) + \nu\sum_{j=1}^{J_m}\gamma_{j_m}\,\mathbb{1}_{\{\mathbf{x}\in R_{jm}\}},$$

where $\nu\in(0,1]$ is the learning rate (shrinkage) and $J_m\le 2^d$ the number of leaves. Smaller ν values lower variance but require more iterations M; deeper trees capture more complex

connections at the cost of interpretability. We tuned the model by performing a grid search over $\nu \in \{0.01, 0.1\}$, $d \in \{1, 3, 5\}$ and $M \in \{100, 300, 500\}$.

Fitted on the **Bush-SMOTE** training set, the tuned GBM yields RMSE = 296, $\boldsymbol{R}^2 = 0.576$.

4.4.8 Extreme Gradient Boosting (XGBoost)

Building on the GBM algorithm, XGBoost retains the same additive model $F_M(\mathbf{x}) = \sum_{m=1}^{M} f_m(\mathbf{x})$ with tree base–learners f_m, however with three main differences (Chen et al., 2015):

- **Second-order optimization:** At iteration m the empirical risk

$$\mathcal{L}(F) = \sum_{i=1}^{n} \ell(y_i, F(\mathbf{x}_i))$$

  is approximated by a second-order Taylor expansion around F_{m-1} given by

$$\tilde{\mathcal{L}}^{(m)} = \sum_{i=1}^{n} \Big[r_{i_m} f_m(\mathbf{x}_i) + \tfrac{1}{2} h_{i_m} f_m(\mathbf{x}_i)^2 \Big] + \Omega(f_m),$$

  where $r_{i_m} = -\partial_F \ell|_{F_{m-1}}$ and $h_{i_m} = \partial_F^2 \ell|_{F_{m-1}}$ are first and second order pseudo-residuals, and $\Omega(f) = \gamma T + \frac{1}{2}\lambda \|\boldsymbol{\omega}\|_2^2$ penalizes the number of leaves T and their scores $\boldsymbol{\omega}$.

- **Regularization:** The term $\Omega(f_m)$ adds a penalty for overly deep or unbalanced trees. When $(\gamma, \lambda) \to \infty$, the model converges, which prevents the variance from getting too big, which can happen in GBM due to large M or tiny learning rates ν.

- **Sampling:** Analogous to Friedman (2001)'s stochastic gradient boosting, XGBoost draws a subsample of observations and predictors for every tree, resulting in lower correlation between the trees and therefore lower variance.

We tuned the model by performing a grid search over

$$\nu \in \{0.05, 0.1\}, \quad d \in \{3, 6\}, \quad \gamma \in \{0, 1\} \quad \text{and} \quad M \in \{200, 500\}.$$

Fitted on the **Bush-SMOTE** training set, the tuned XGBoost algorithm yields RMSE = 313, $\boldsymbol{R}^2 = 0.624$; the highest explanatory power overall.

4.4.9 Feed-Forward Neural Networks

A feed-forward neural network, also called a multilayer perceptron (MLP), stacks L hidden layers, each applying an affine map followed by a non-linear activation

$$\mathbf{a}^{(\ell)} = \sigma\big(\mathbf{W}^{(\ell)} \mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)}\big), \qquad \mathbf{a}^{(0)} = \mathbf{x}, \quad \ell = 1, \ldots, L,$$

and finishes with a linear output node for regression $\hat{\mathbf{y}} = \mathbf{a}^{(L+1)}$ (Bebis and Georgiopoulos, 2002). According to Hornik (1991), multilayer feed-forward neural networks are, under very general conditions on the hidden unit activation function and given that sufficiently many hidden units are available, universal approximators. The parameters $\Theta = \{\mathbf{W}^{(\ell)}, \mathbf{b}^{(\ell)}\}_{\ell=1}^{L}$ minimize the mean–squared error $\mathcal{L}(\Theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\mathbf{y}}_i)^2$ through

back-propagation. We trained the network by centering and scaling the inputs, performed a Yeo-Johnson transformation (Weisberg, 2001) and applied PCA, retaining at most ten components. Furthermore, a random grid of $n = 25$ candidate networks searched hidden units $s \in \{1, \dots, 8\}$ and decay $\lambda \in 10^{[-5,-1]}$.

Fitted on the **Bush-SMOTE** training set, the tuned neural network yields RMSE = 462, $\boldsymbol{R}^2 = 0.485$.

4.4.10 Bagged Neural Networks

Bootstrap aggregation can be applied in order to reduce overfitting and the high variance of single MLPs (Ha et al., 2005). From the original data we drew B bootstrap samples, trained one independent 1-hidden-layer network on each, and averaged their predictions,

$$\hat{f}_{\text{bag}}(\mathbf{x}) \;=\; \frac{1}{B}\sum_{b=1}^{B} f_{\text{MLP}}^{(b)}(\mathbf{x}),$$

which lowers variance while keeping bias essentially unchanged (Breiman, 1996). We trained the network by centering and scaling the inputs, performed a Yeo-Johnson transformation and applied PCA, retaining at most ten components. Furthermore, a random grid of $n = 25$ candidate networks searched hidden units $s \in \{1, \dots, 6\}$ and decay $\lambda \in 10^{[-5,-2]}$ while setting the number of repeats to $B = 30$.

Fitted on the **Bush-SMOTE** training set, the bagged neural network attains RMSE = 396, $\boldsymbol{R}^2 = 0.472$, showing that bagging reduces variance relative to a single MLP yet still lags behind tree-based boosting.

Chapter 5

Conclusion and Future Work

Conclusion

We have constructed a performance metric for the ranking of Swiss elite climbers across an entire season, by comparing five ranking methods. We compared different supervised learning models under data scarcity using two virtual sample generation (VSG) methods. We were able to construct a model which can predict with a for practice meaningful accuracy the seasonal performance of a climbing athlete given a collection of attributes of the climber. This shows the potential of data science in the field of sport analytics, where limited data availability still persists and small improvements can make big differences.

For the performance score, the adapted IFSC-based score provides an accurate, reproducible baseline that treats international and national events coherently. Section 2.4 compares the adapted IFSC, PageRank, Massey, Colley and Elo ranking methods (see Figures 2.3 and 2.4, Table 2.2). The adapted IFSC-based score is preferred because it respects the all-against-field nature of climbing rather than head to head encounters. It aligns closely with PageRank at the top of the table (Figures 2.3 and 2.4), yet avoids pairwise graph construction, which reduces complexity and dependence on other athletes' participation patterns.

Across the supervised learning models, tree-based models trained with the Bush–SMOTE augmented dataset have the best prediction accuracy on the 2024 testing dataset (Figure 4.1, Table 4.4). Using Bush-SMOTE VSG with gradient boosting reduces the RMSE by roughly 20% and increases $\boldsymbol{R}^2$ from 0.18 to 0.58 compared to gradient boosting with the non-augmented dataset, while XGBoost attains the highest $\boldsymbol{R}^2$ of 0.62. Tree-based models likely benefit from the Bush-SMOTE VSG due to the increased possibilities of tree splits on the augmented dataset. Given the geometrical nature of the Bush-SMOTE, decision trees can better refine their decision, making the models more accurate by lowering the validation error. In contrast, distance-based models (kNN, Ridge, SVM, Lasso and OLS) benefit more from GMM-VSG, which generates cluster-consistent neighborhoods (amplifying the underlying distribution) by sampling from fitted mixture components. This results in improved RMSE compared to the base or Bush-SMOTE datasets (Table 4.4). Simple neural networks do not consistently gain from augmentation, which is likely since the added variability raises variance more than it reduces the bias in this model.

One should note that VSG remains somewhat unpopular in statistics, because it ignores independence of observations, normality and independent errors. These are important

assumptions when the goal is inference (Liese and Miescke, 2008). Nevertheless, the field of materials science does rely on VSG (Wen et al., 2024) because in this context observations are limited, given their costly collection process, and the goal is prediction rather than inference. Domain boundaries, which are often known in this field (e.g., physically plausible ranges), further regularize sampling.

Future Work

The training dataset we used is small, single-season, and national-team specific. Future work may analyze multi-season and international data and compare the findings to this work. This might be especially interesting since climbing competitions are constantly evolving with rising popularity of the sport. Psychological measurements collected across multiple seasons would also offer different insights to further train and evaluate supervised learning models. In this work, we only compared sport ranking methods which rely on the outcome of single games. An interesting ranking method for future work is provided by what is called social choice theory. Smead (2019), who applied social choice theory to the context of sports tournaments, suggests that ranking methods should handle the case of cycles, such as A beats B, B beats C, C beats A.

Other work may be to further test the limitations and the potential of VSG:

- Estimate a learning curve on progressively larger base training datasets. This would quantify when VSG becomes negligible. One could implement this by choosing different sizes of sub-samples from a big known dataset.
- Estimate a learning curve on progressively larger augmented datasets. By testing different numbers of virtually generated observations one could find the ideal ratio of base training dataset size vs. augmented dataset size.
- Compare the performance of VSG on different, known data distributions. Does VSG perform better given specific distributions, e.g. normal or exponential?
- Estimate the noise sensitivity. By injecting noise into the targets one could measure the performance degradation with and without VSG in order to test the robustness of VSG.
- Assess the bias–variance decomposition. Using repeated CV, one could decompose the error to see whether VSG primarily reduces variance or introduces bias for each model.

Bibliography

Achar, S. K. and J. A. Keith (2024). Small data machine learning approaches in molecular and materials science. *Chemical Reviews 124*(24), 13571–13573.

Awad, M. and R. Khanna (2015). Support vector regression. In *Efficient learning machines: Theories, concepts, and applications for engineers and system designers*, pp. 67–80. Springer.

Bebis, G. and M. Georgiopoulos (2002). Feed-forward neural networks. *Ieee Potentials 13*(4), 27–31.

Bera, A. K. and Y. Bilias (2002). The mm, me, ml, el, ef and gmm approaches to estimation: a synthesis. *Journal of Econometrics 107*(1-2), 51–86.

Biernacki, C., G. Celeux, and G. Govaert (2003). Choosing starting values for the em algorithm for getting the highest likelihood in multivariate gaussian mixture models. *Computational Statistics & Data Analysis 41*(3-4), 561–575.

Breiman, L. (1996). Bagging predictors. *Machine learning 24*(2), 123–140.

Breiman, L. (2001). Random forests. *Machine learning 45*(1), 5–32.

Chartier, T. P., E. Kreutzer, A. N. Langville, and K. E. Pedings (2011). Sports ranking with nonuniform weighting. *Journal of Quantitative Analysis in Sports 7*(3).

Chawla, N. V., K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer (2002). Smote: synthetic minority over-sampling technique. *Journal of artificial intelligence research 16*, 321–357.

Chen, T., T. He, M. Benesty, V. Khotilovich, Y. Tang, H. Cho, K. Chen, R. Mitchell, I. Cano, T. Zhou, et al. (2015). Xgboost: extreme gradient boosting. *R package version 0.4-2 1*(4), 1–4.

Chen, Z., Z. Lv, R. Di, P. Wang, X. Li, X. Sun, and Y. Xu (2023). A novel virtual sample generation method to improve the quality of data and the accuracy of data-driven models. *Neurocomputing 548*, 126380.

Chen, Z.-S., Q.-X. Zhu, Y. Xu, Y.-L. He, Q.-L. Su, Y. C. Liu, and Z. K. Nagy (2021). Integrating virtual sample generation with input-training neural network for solving small sample size problems: application to purified terephthalic acid solvent system. *Soft Computing 25*, 6489–6504.

Chipman, J. S. (2011). Gauss-markov theorem. In *International Encyclopedia of Statistical Science*, pp. 577–582. Springer.

Chmielewska, A. and B. Regulska-Ilow (2023). The evaluation of energy availability and

dietary nutrient intake of sport climbers at different climbing levels. *International Journal of Environmental Research and Public Health 20*(6), 5176.

Colley, W. (2002). Colley's bias free college football ranking method. *Princeton, NJ, USA: Princeton University*.

Cortes, C. and V. Vapnik (1995). Support-vector networks. *Machine learning 20*(3), 273–297.

Dou, B., Z. Zhu, E. Merkurjev, L. Ke, L. Chen, J. Jiang, Y. Zhu, J. Liu, B. Zhang, and G.-W. Wei (2023). Machine learning methods for small data challenges in molecular science. *Chemical Reviews 123*(13), 8736–8780.

Elo, A. E. (1978). The rating of chessplayers: Past and present. *(No Title)*.

Elreedy, D. and A. F. Atiya (2019). A comprehensive analysis of synthetic minority oversampling technique (smote) for handling class imbalance. *Information Sciences 505*, 32–64.

Fan, W. and A. Bifet (2013). Mining big data: current status, and forecast to the future. *ACM SIGKDD explorations newsletter 14*(2), 1–5.

Fidora, A. and C. Sierra (2011). *Ramon Llull, from the Ars Magna to artificial intelligence*. Artificial Intelligence Research Institute Barcelona.

Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of statistics*, 1189–1232.

Gong, H.-F., Z.-S. Chen, Q.-X. Zhu, and Y.-L. He (2017). A monte carlo and pso based virtual sample generation method for enhancing the energy prediction and energy optimization on small data problem: An empirical study of petrochemical industries. *Applied Energy 197*, 405–415.

Ha, K., S. Cho, and D. MacLachlan (2005). Response models based on bagging neural networks. *Journal of Interactive Marketing 19*(1), 17–30.

Hoerl, R. W. (2020). Ridge regression: a historical context. *Technometrics 62*(4), 420–425.

Hornik, K. (1991). Approximation capabilities of multilayer feedforward networks. *Neural networks 4*(2), 251–257.

IFSC (2024). Ifsc results. https://ifsc.results.info.

IFSC (2025a). IFSC World Rankings 2025; Counting competitions and calculation factors. https://images.ifsc-climbing.org/ifsc/image/private/t_q_good/prd/eunqmv492fipy6gi7zth.pdf.

IFSC (2025b). National Federations Quotas: 2025 IFSC season. https://images.ifsc-climbing.org/ifsc/image/private/t_q_good/prd/osiy36gs41ctolsw99ns.pdf.

IFSC (2025c). Team World Rankings. https://www.ifsc-climbing.org/rankings/team.

IFSC (2025d). World Rankings; Regulations relating to the world ranking system. https://images.ifsc-climbing.org/ifsc/image/private/t_q_good/prd/hrw5bjmok8vmypruobxf.pdf.

Jordan, M. I. and T. M. Mitchell (2015). Machine learning: Trends, perspectives, and prospects. *Science 349*(6245), 255–260.

Kramer, O. (2013). K-nearest neighbors. In *Dimensionality reduction with unsupervised nearest neighbors*, pp. 13–23. Springer.

Langville, A. N. and C. D. Meyer (2006). *Google's PageRank and beyond: The science of search engine rankings*. Princeton university press.

Langville, A. N. and C. D. Meyer (2012). *Who's# 1? The science of rating and ranking*. Princeton University Press.

Lawrence, C. and W. J. Krzanowski (1996). Mixture separation for mixed-mode data. *Statistics and Computing 6*, 85–92.

Li, L., S. K. Damarla, Y. Wang, and B. Huang (2021). A gaussian mixture model based virtual sample generation approach for small datasets in industrial processes. *Information Sciences 581*, 262–277.

Liese, F. and K.-J. Miescke (2008). Statistical decision theory. In *Statistical Decision Theory: Estimation, Testing, and Selection*, pp. 1–52. Springer.

Lower, L. M. and W. A. Czekanski (2019). Effective management of scarce resources: A case study of american collegiate sport clubs. *Managing Sport and Leisure 24*(1-3), 119–140.

Luo, H. and S. G. Paal (2021). Reducing the effect of sample bias for small data sets with double-weighted support vector transfer regression. *Computer-Aided Civil and Infrastructure Engineering 36*(3), 248–263.

Maimon, O. and L. Rokach (2007). *Soft computing for knowledge discovery and data mining*. Springer Science & Business Media.

Markoff, J. (2005). Debating the size of the web. *International Herald Tribune 16*.

Massey, K. (1997). Statistical models applied to the rating of sports teams. *Bluefield College 1077*.

McLachlan, G. J. and K. E. Basford (1988). *Mixture models: Inference and applications to clustering*, Volume 38. M. Dekker New York.

Moon, T. K. (1996). The expectation-maximization algorithm. *IEEE Signal processing magazine 13*(6), 47–60.

Ochieng, P. J., A. London, and M. Krész (2022). A forward-looking approach to compare ranking methods for sports. *Information 13*(5).

Pais, D., S. Brás, and R. Sebastião (2024). Overcoming the small dataset challenge in healthcare. In *2024 IEEE 22nd Mediterranean Electrotechnical Conference (MELECON)*, pp. 497–502. IEEE.

Pelayo, L. and S. Dick (2019). Synthetic minority oversampling for function approximation problems. *International Journal of Intelligent Systems 34*(11), 2741–2768.

Ranstam, J. and J. A. Cook (2018). Lasso regression. *Journal of British Surgery 105*(10), 1348–1348.

Reynolds, D. A. et al. (2009). Gaussian mixture models. *Encyclopedia of biometrics 741*(659-663), 3.

SAC-CAS (2024). Sac cas results. https://sac-cas.results.info.

Shen, L. and Q. Qian (2022). A virtual sample generation algorithm supporting machine learning with a small-sample dataset: A case study for rubber materials. *Computational Materials Science 211*, 111475.

Smead, R. (2019). Sports tournaments and social choice theory. *Philosophies 4*, 28.

Sutojo, T., S. Rustad, M. Akrom, A. Syukur, G. F. Shidik, and H. K. Dipojono (2023). A machine learning approach for corrosion small datasets. *npj materials degradation 7*(1), 18.

Sutojo, T., A. Syukur, S. Rustad, G. F. Shidik, H. A. Santoso, P. Purwanto, and M. Muljono (2020). Investigating the impact of synthetic data distribution on the performance of regression models to overcome small dataset problems. In *2020 International Seminar on Application for Technology of Information and Communication (iSemantic)*, pp. 125–130. IEEE.

Tan, A. (2025). The effect of nutrition on athletic performance with a focus on rock climbing. *The Lens Journal*.

Tao, X., Q. Li, W. Guo, C. Ren, Q. He, R. Liu, and J. Zou (2020). Adaptive weighted over-sampling for imbalanced datasets based on density peaks clustering with heuristic filtering. *Information Sciences 519*, 43–73.

Weisberg, S. (2001). Yeo-johnson power transformations. *Department of Applied Statistics, University of Minnesota. Retrieved June 1*(2003), 8.

Wen, J., A. Su, X. Wang, H. Xu, J. Ma, K. Chen, X. Ge, Z. Xu, and Z. Lv (2024). Virtual sample generation for small sample learning: a survey, recent developments and future prospects. *Neurocomputing*, 128934.

Xu, P., X. Ji, M. Li, and W. Lu (2023). Virtual sample generation in machine learning assisted materials design and discovery. *Journal of Materials Informatics 3*(3), N–A.

Xu, Q.-S. and Y.-Z. Liang (2001). Monte carlo cross validation. *Chemometrics and Intelligent Laboratory Systems 56*(1), 1–11.

Yang, J., X. Yu, Z.-Q. Xie, and J.-P. Zhang (2011). A novel virtual sample generation method based on gaussian distribution. *Knowledge-Based Systems 24*(6), 740–748.

Appendix A

Data

A.1 Variable Overview

A.1.1 Physiological Variables

Variable Name	Description	Unit	Type
Height_cm	Athlete height	cm	Continuous
BW_kg	Athlete weight	kg	Continuous
BW_N	Athlete weight	N	Continuous
FS_MVC_L_N	Max. finger strength pull force left	N	Continuous
FS_MVC_R_N	Max. finger strength pull force right	N	Continuous
Hip_split_cm	Span of legs in max. split position	cm	Continuous
Hip_open_cm	Distance to ground when opening the hips while laying on back	cm	Continuous
Foot_raise_R_cm	Distance to ground when raising right foot	cm	Continuous
Foot_raise_L_cm	Distance to ground when raising left foot	cm	Continuous
PullUp_MW_kg	Max. additional weight while still completing a pull up	kg	Continuous
Slap_R_cm	Climbing SLAP test right hand	cm	Continuous
Slap_L_cm	Climbing SLAP test right hand	cm	Continuous
Gender	Athlete gender	M / F	Nominal

A.1.2 Psychological Variables

Variable Name	Description	Unit	Type
Training_D	Number of training days	Days	Discrete
FullRest_D	Number of full rest days	Days	Discrete
TS_Frust_15	Usual frustration during training sessions	1 - 5	Ordinal
TS_Satis_15	Usual satisfaction after training session	1 - 5	Ordinal
Comp_Comfort_15	Usual comfort level during competitions	1 - 5	Ordinal
Performance_better	Evaluation of performance in training vs. competition	T / C	Nominal
Preference_TCB	Training vs. competition (or both) preference	T / C / B	Nominal
AVG_sleep_hrs	Average sleep per night	Hours	Discrete
AVG_sleep_comp_hrs	Average sleep per night before competitions	Hours	Discrete
Diet	Athlete follows some form of diet	Yes / No	Nominal
Climbing_Life	How much life revolves around climbing	1 - 5	Ordinal
Team_comfort_15	Comfort feeling within the team	1 - 5	Ordinal
Team_vibe_15	Perceived general vibes within the team	1 - 5	Ordinal
Goals	Presence of Goals	Yes / No	Nominal
Goals_sup_coach_15	Perceived support of the SCT coaches	1 - 5	Ordinal
Goals_archieve_15	Achievement of individual goals	1 - 5	Ordinal
Main_coach	Main Coach (SCT or Ext.)	SCT / E	Nominal
Satisf_SCTCoach_15	Satisfaction with SCT Coaches	1 - 5	Ordinal
Satisf_SCT_15	General satisfaction with the Swiss Climbing Team	1 - 5	Ordinal

A.1.3 Outcome Variable

Variable Name	Description	Unit	Type
IFSC.Score	Athletes' Performance Score from Chapter 2	points	Continuous

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies. **In consultation with the supervisor**, one of the following two options must be selected:

☐ I hereby declare that I authored the work in question independently, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies[1].

☒ I hereby declare that I authored the work in question independently. In doing so I only used the authorised aids, which included suggestions from the supervisor regarding language and content and generative artificial intelligence technologies. The use of the latter and the respective source declarations proceeded in consultation with the supervisor.

Title of paper or thesis:

Managing Data Scarcity in Sports Analytics: Oversampling-Assisted Machine Learning for Forecasting Swiss Elite Climber Results

Authored by:
If the work was compiled in a group, the names of all authors are required.

Last name(s):	**First name(s):**
Eckmann	Anthony

With my signature I confirm the following:
- I have adhered to the rules set out in the Citation Guidelines.
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

Place, date	**Signature(s)**
Zurich, 26.08.2025	

If the work was compiled in a group, the names of all authors are required. Through their signatures they vouch jointly for the entire content of the written work.

[1] For further information please consult the ETH Zurich websites, e.g. https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html and https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html (subject to change).